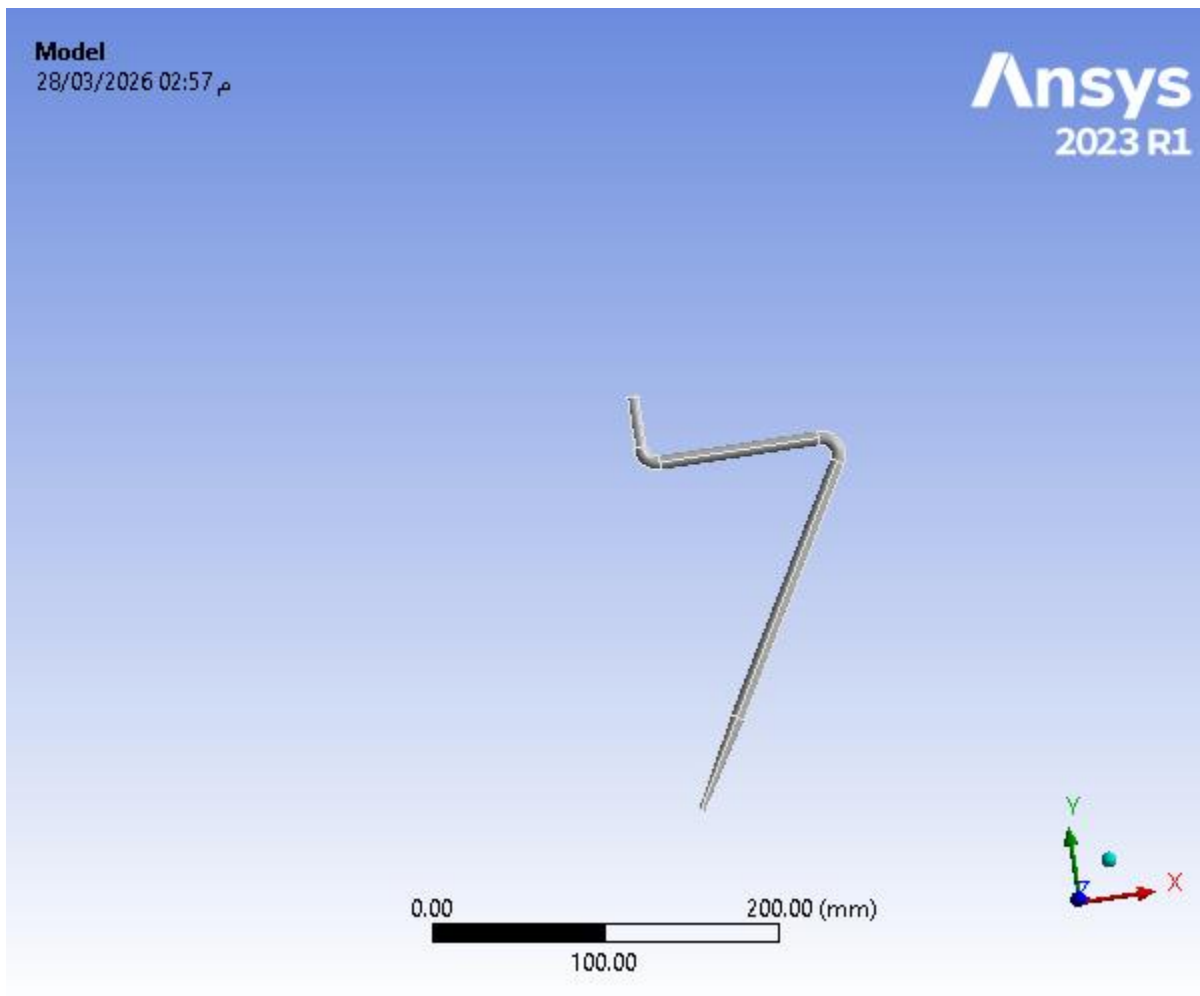




Project*

First Saved	Tuesday, February 24, 2026
Last Saved	Tuesday, February 24, 2026
Product Version	2023 R1
Save Project Before Solution	No
Save Project After Solution	No



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Report Not Finalized

Not all objects described below are in a finalized state. As a result, data may be incomplete, obsolete or in error. [View first state problem](#). To finalize this report, edit objects as needed and solve the analyses.

Units

TABLE 1

Unit System	Metric (mm, kg, N, s, mV, mA) Degrees rad/s Celsius
Angle	Degrees
Rotational Velocity	rad/s
Temperature	Celsius

Model (A4)

TABLE 2

Model (A4) > Geometry Imports

Object Name	Geometry Imports
State	Solved

TABLE 3
Model (A4) > Geometry Imports > Geometry Import (A3)

Object Name	<i>Geometry Import (A3)</i>
State	Solved
Definition	
Source	C:\Users\user1\Desktop\blades\blenderaugerlong.STEP
Type	Step
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	No
Parameters	Independent
Parameter Key	ANS;DS
Attributes	No
Named Selections	No
Material Properties	No
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	No
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
Enclosure and Symmetry Processing	Yes

Geometry

TABLE 4
Model (A4) > Geometry

Object Name	<i>Geometry</i>
State	Fully Defined
Definition	
Source	C:\Users\user1\Desktop\blades\blenderaugerlong.STEP
Type	Step
Length Unit	Millimeters
Element Control	Program Controlled
Display Style	Body Color
Bounding Box	
Length X	120.68 mm
Length Y	243.4 mm
Length Z	8. mm
Properties	
Volume	17061 mm ³

Mass	0.13393 kg
Scale Factor Value	1.
Statistics	
Bodies	1
Active Bodies	1
Nodes	110070
Elements	69652
Mesh Metric	Skewness
Min	2.0742e-003
Max	0.88539
Average	0.30349
Standard Deviation	0.14145
Update Options	
Assign Default Material	No
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	No
Parameters	Independent
Parameter Key	ANS;DS
Attributes	No
Named Selections	No
Material Properties	No
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	No
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
Enclosure and Symmetry Processing	Yes

TABLE 5
Model (A4) > Geometry > Parts

Object Name	<i>blenderaugerlong-FreeParts Split Line1</i>
State	Meshed
Graphics Properties	
Visible	Yes
Transparency	1
Definition	
Suppressed	No
Stiffness Behavior	Flexible
Coordinate System	Default Coordinate System

Reference Temperature	By Environment
Treatment	None
Material	
Assignment	Structural Steel
Nonlinear Effects	Yes
Thermal Strain Effects	Yes
Bounding Box	
Length X	120.68 mm
Length Y	243.4 mm
Length Z	8. mm
Properties	
Volume	17061 mm ³
Mass	0.13393 kg
Centroid X	57.317 mm
Centroid Y	-54.433 mm
Centroid Z	-8.9487e-005 mm
Moment of Inertia Ip1	542.83 kg·mm ²
Moment of Inertia Ip2	150.93 kg·mm ²
Moment of Inertia Ip3	692.71 kg·mm ²
Statistics	
Nodes	110070
Elements	69652
Mesh Metric	Skewness
Min	2.0742e-003
Max	0.88539
Average	0.30349
Standard Deviation	0.14145

TABLE 6
Model (A4) > Materials

Object Name	<i>Materials</i>
State	Fully Defined
Statistics	
Materials	2
Material Assignments	0

Coordinate Systems

TABLE 7
Model (A4) > Coordinate Systems > Coordinate System

Object Name	<i>Global Coordinate System</i>
State	Fully Defined
Definition	
Type	Cartesian
Coordinate System ID	0.
Origin	
Origin X	0. mm
Origin Y	0. mm

Origin Z	0. mm
Directional Vectors	
X Axis Data	[1. 0. 0.]
Y Axis Data	[0. 1. 0.]
Z Axis Data	[0. 0. 1.]

Mesh

TABLE 8
Model (A4) > Mesh

Object Name	<i>Mesh</i>
State	Solved
Display	
Display Style	Use Geometry Setting
Defaults	
Physics Preference	Mechanical
Element Order	Program Controlled
Element Size	1.0 mm
Sizing	
Use Adaptive Sizing	Yes
Resolution	7
Mesh Defeaturing	Yes
Defeature Size	Default
Transition	Slow
Span Angle Center	Medium
Initial Size Seed	Part
Bounding Box Diagonal	271.79 mm
Average Surface Area	367.92 mm ²
Minimum Edge Length	2.8909 mm
Quality	
Check Mesh Quality	Yes, Errors
Error Limits	Aggressive Mechanical
Target Element Quality	0.95
Smoothing	Medium
Mesh Metric	Skewness
Min	2.0742e-003
Max	0.88539
Average	0.30349
Standard Deviation	0.14145
Inflation	
Use Automatic Inflation	None
Inflation Option	Smooth Transition
Transition Ratio	0.272
Maximum Layers	5
Growth Rate	1.2
Inflation Algorithm	Pre
View Advanced Options	No
Advanced	

Number of CPUs for Parallel Part Meshing	Program Controlled
Straight Sided Elements	No
Rigid Body Behavior	Dimensionally Reduced
Triangle Surface Mesher	Program Controlled
Topology Checking	Yes
Pinch Tolerance	Please Define
Generate Pinch on Refresh	No
Statistics	
Nodes	110070
Elements	69652
Show Detailed Statistics	No

TABLE 9
Model (A4) > Mesh > Mesh Controls

Object Name	<i>Face Meshing</i>
State	Ignored
Scope	
Scoping Method	Geometry Selection
Geometry	25 Faces
Definition	
Suppressed	No
Mapped Mesh	Yes
Constrain Boundary	No
Advanced	
Specified Sides	No Selection
Specified Corners	No Selection
Specified Ends	No Selection

Static Structural (A5)

TABLE 10
Model (A4) > Analysis

Object Name	<i>Static Structural (A5)</i>
State	Solved
Definition	
Physics Type	Structural
Analysis Type	Static Structural
Solver Target	Mechanical APDL
Options	
Environment Temperature	22. °C
Generate Input Only	No

TABLE 11
Model (A4) > Static Structural (A5) > Analysis Settings

Object Name	<i>Analysis Settings</i>
State	Fully Defined
Step Controls	
Number Of Steps	1.

Current Step Number	1.
Step End Time	1. s
Auto Time Stepping	Program Controlled
Solver Controls	
Solver Type	Program Controlled
Weak Springs	Off
Solver Pivot Checking	Program Controlled
Large Deflection	Off
Inertia Relief	Off
Quasi-Static Solution	Off
Rotordynamics Controls	
Coriolis Effect	Off
Restart Controls	
Generate Restart Points	Program Controlled
Retain Files After Full Solve	No
Combine Restart Files	Program Controlled
Nonlinear Controls	
Newton-Raphson Option	Program Controlled
Force Convergence	Program Controlled
Moment Convergence	Program Controlled
Displacement Convergence	Program Controlled
Rotation Convergence	Program Controlled
Line Search	Program Controlled
Stabilization	Program Controlled
Advanced	
Inverse Option	No
Contact Split (DMP)	Off
Output Controls	
Stress	Yes
Back Stress	No
Strain	Yes
Contact Data	Yes
Nonlinear Data	No
Nodal Forces	No
Volume and Energy	Yes
Euler Angles	Yes
General Miscellaneous	No
Contact Miscellaneous	No
Store Results At	All Time Points
Result File Compression	Program Controlled
Analysis Data Management	
Solver Files Directory	F:\blades\longblades_files\ldp0\SYS\MECH\
Future Analysis	None
Scratch Solver Files Directory	
Save MAPDL db	No
Contact Summary	Program Controlled
Delete Unneeded Files	Yes
Nonlinear Solution	No

Solver Units	Active System
Solver Unit System	nmm

TABLE 12
Model (A4) > Static Structural (A5) > Loads

Object Name	Fixed Support	Pressure	Moment
State	Fully Defined		
Scope			
Scoping Method	Geometry Selection		
Geometry	1 Face	25 Faces	10 Faces
Definition			
Type	Fixed Support	Pressure	Moment
Suppressed	No		
Define By		Components	
Applied By		Surface Effect	
Loaded Area		Deformed	
Coordinate System		Global Coordinate System	
X Component		0. MPa (ramped)	0. N·mm (ramped)
Y Component		-4.3e-003 MPa (ramped)	-740. N·mm (ramped)
Z Component		0. MPa (ramped)	0. N·mm (ramped)
Behavior			Deformable
Advanced			
Pinball Region			All

FIGURE 1
Model (A4) > Static Structural (A5) > Fixed Support > Figure

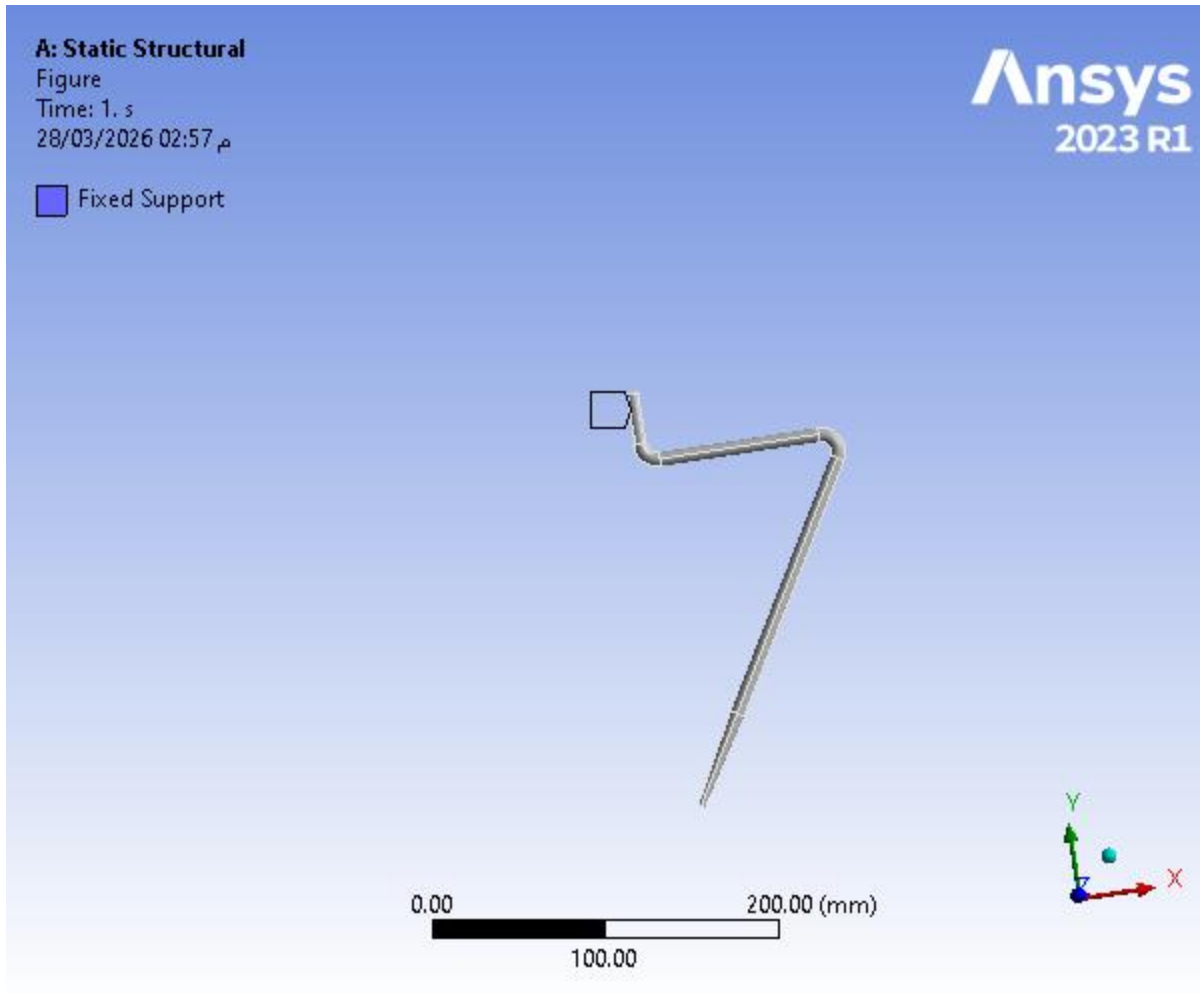


FIGURE 2
Model (A4) > Static Structural (A5) > Fixed Support > Image

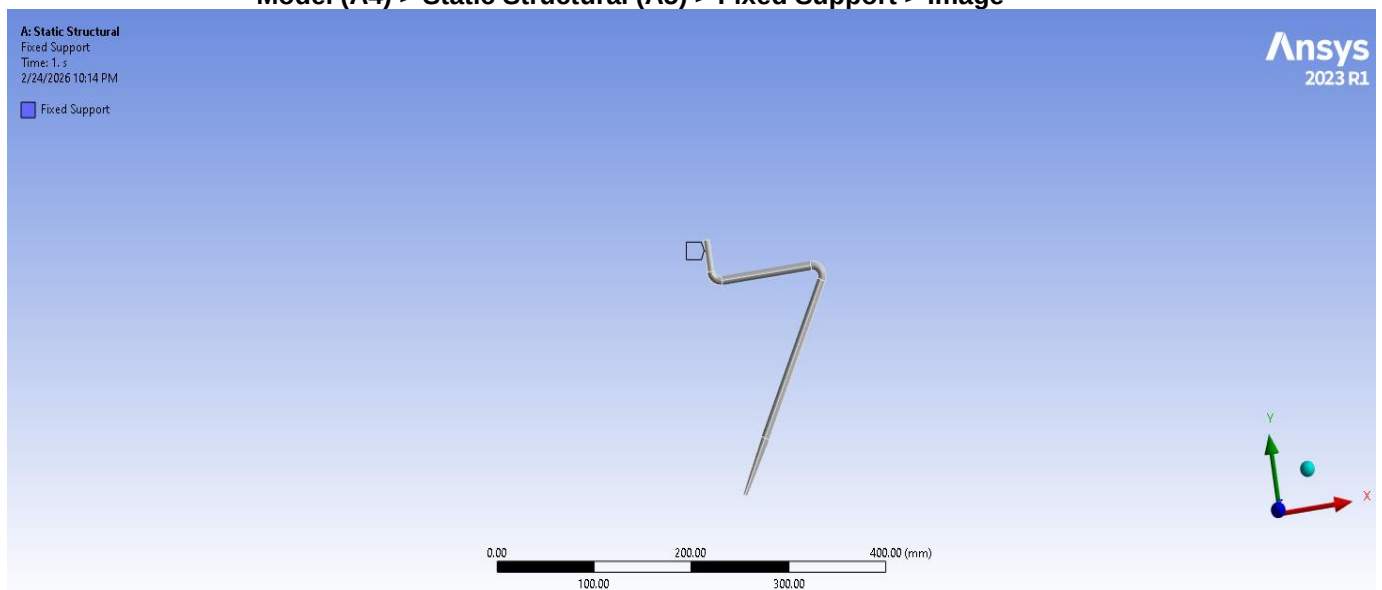


FIGURE 3
Model (A4) > Static Structural (A5) > Pressure

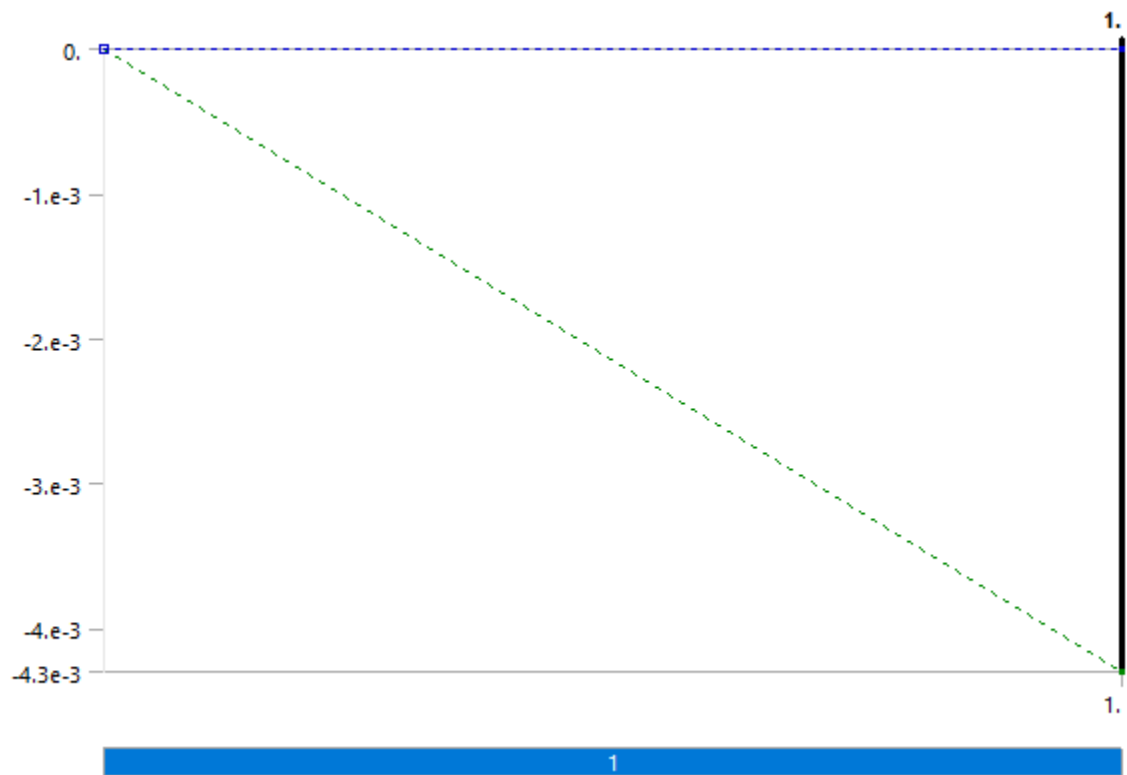


FIGURE 4
Model (A4) > Static Structural (A5) > Pressure > Figure

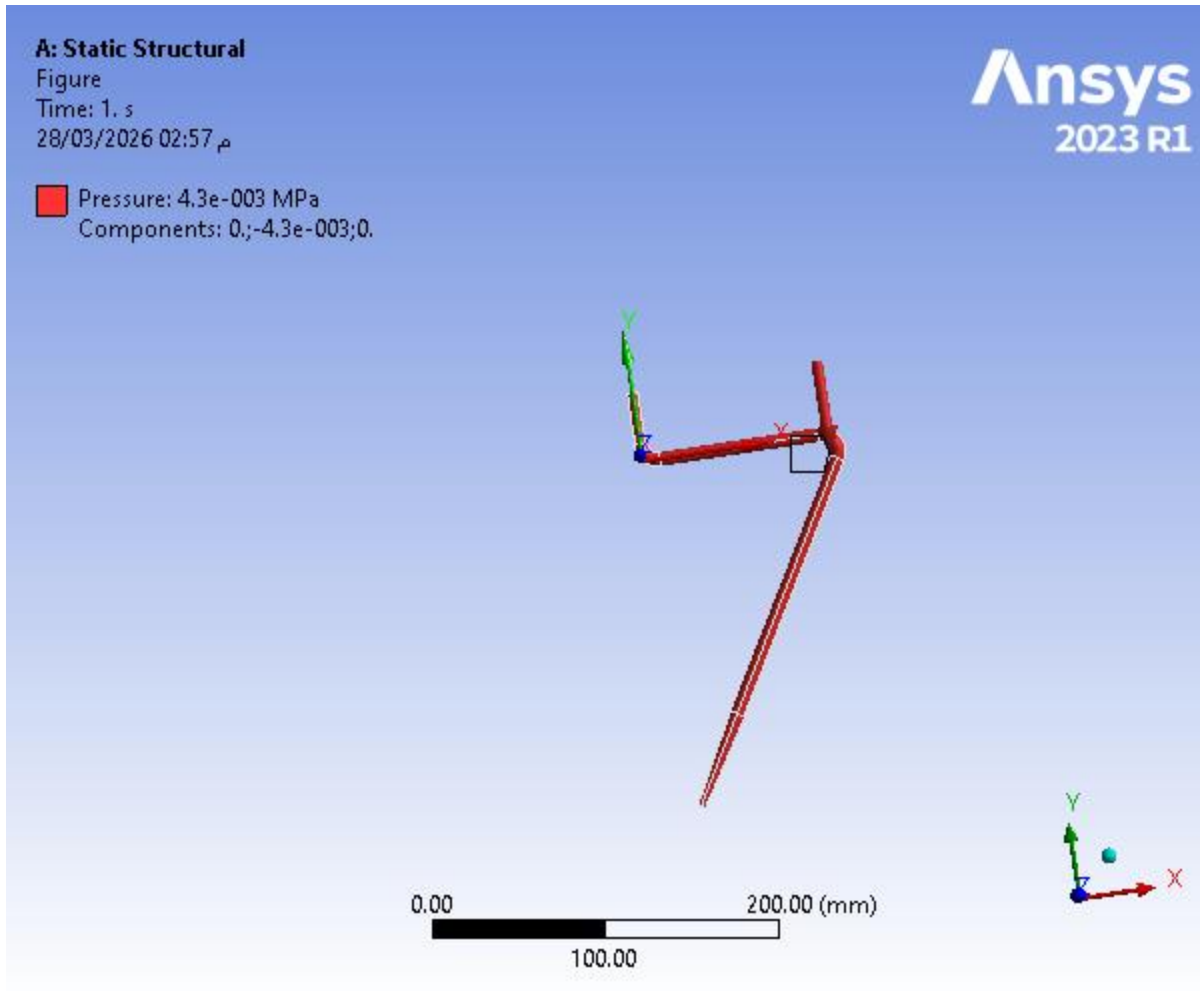


FIGURE 5
Model (A4) > Static Structural (A5) > Pressure > Image

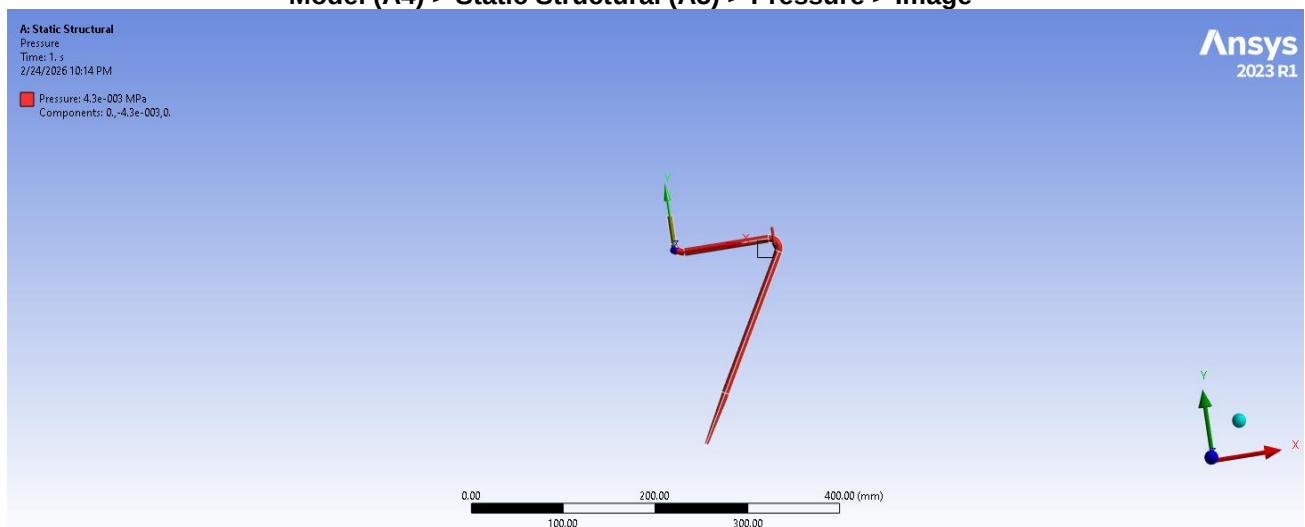


FIGURE 6
Model (A4) > Static Structural (A5) > Moment

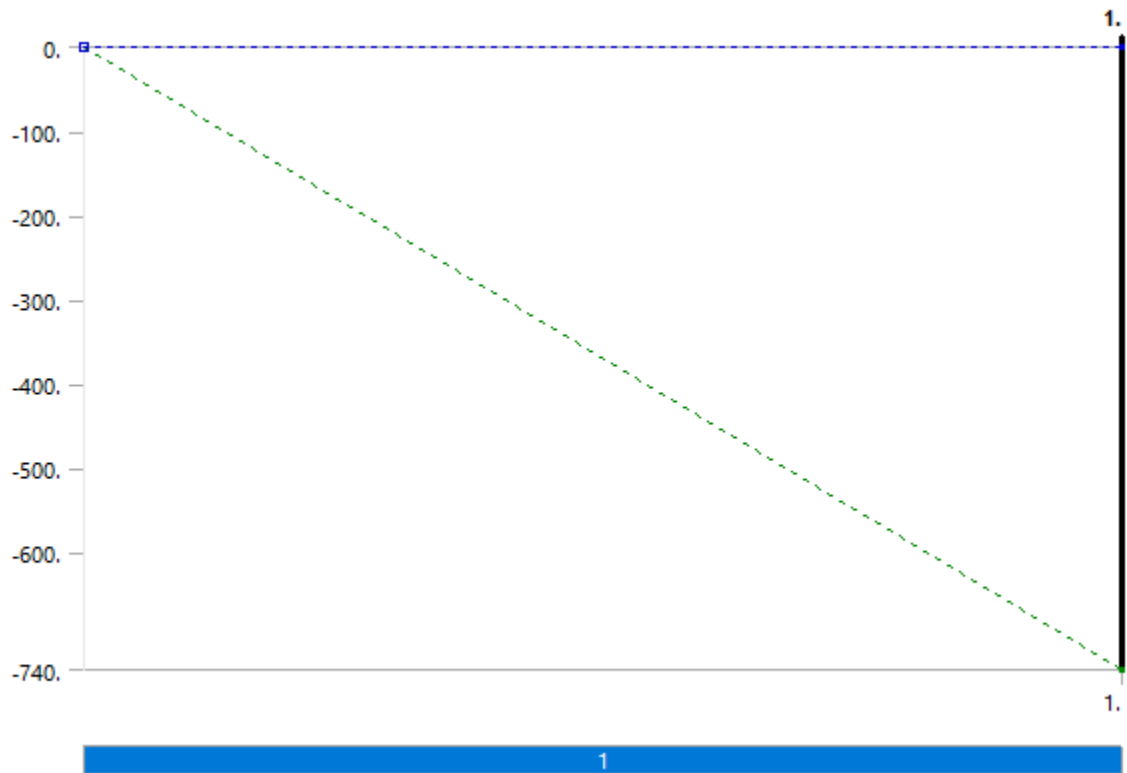


FIGURE 7
Model (A4) > Static Structural (A5) > Moment > Figure

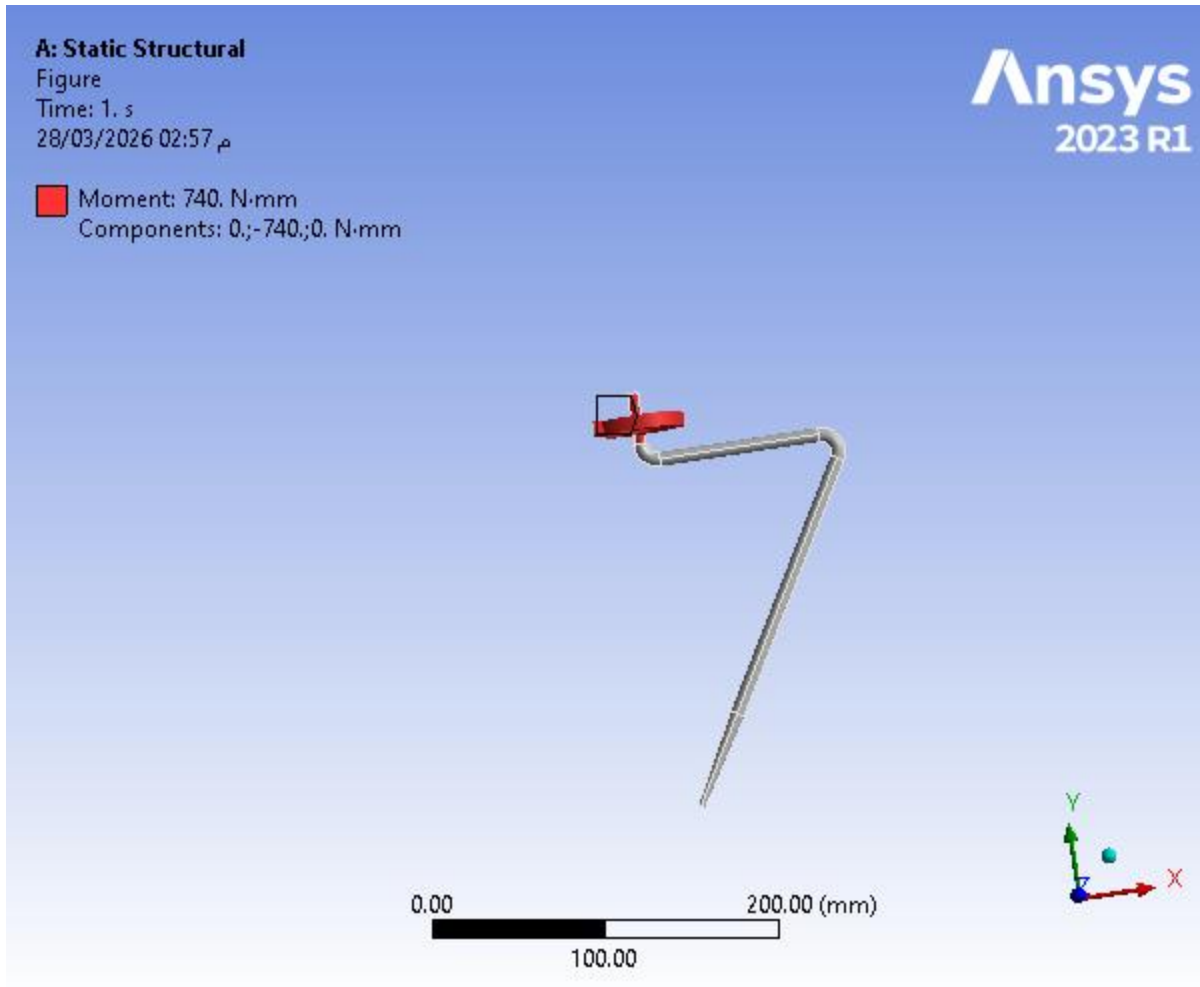
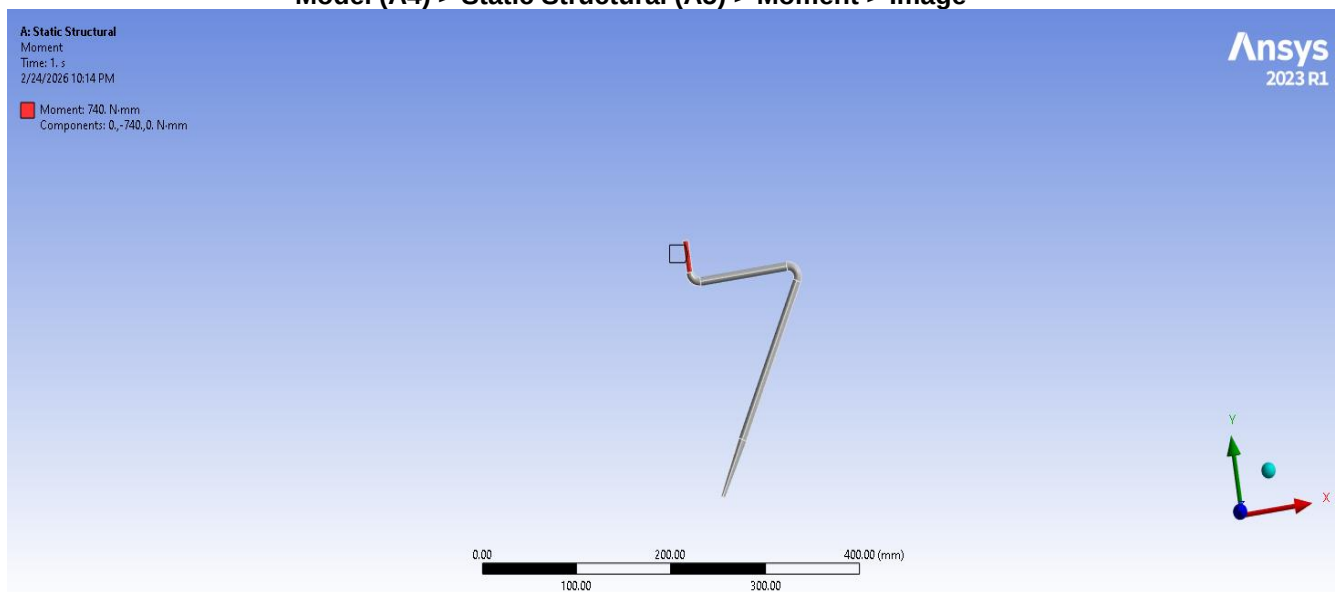


FIGURE 8
Model (A4) > Static Structural (A5) > Moment > Image



Solution (A6)

TABLE 13
Model (A4) > Static Structural (A5) > Solution

Object Name	<i>Solution (A6)</i>
State	Solved
Adaptive Mesh Refinement	
Max Refinement Loops	1.
Refinement Depth	2.
Information	
Status	Done
MAPDL Elapsed Time	11. s
MAPDL Memory Used	633. MB
MAPDL Result File Size	47.875 MB
Post Processing	
Beam Section Results	No
On Demand Stress/Strain	No

TABLE 14
Model (A4) > Static Structural (A5) > Solution (A6) > Solution Information

Object Name	<i>Solution Information</i>
State	Solved
Solution Information	
Solution Output	Solver Output
Newton-Raphson Residuals	0
Identify Element Violations	0
Update Interval	2.5 s
Display Points	All
FE Connection Visibility	
Activate Visibility	Yes
Display	All FE Connectors
Draw Connections Attached To	All Nodes
Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines

TABLE 15
Model (A4) > Static Structural (A5) > Solution (A6) > Results

Object Name	Total Deformation	Maximum Shear Stress	Equivalent Stress
State	Solved		
Scope			
Scoping Method	Geometry Selection		
Geometry	All Bodies		
Definition			
Type	Total Deformation	Maximum Shear Stress	Equivalent (von-Mises) Stress
By	Time		
Display Time	Last		
Separate Data by Entity	No		

Calculate Time History	Yes		
Identifier			
Suppressed	No		
Results			
Minimum	0. mm	3.992e-003 MPa	7.3206e-003 MPa
Maximum	0.23247 mm	42.824 MPa	85.366 MPa
Average	0.15679 mm	3.1889 MPa	6.1837 MPa
Minimum Occurs On	blenderaugerlong-FreeParts Split Line1		
Maximum Occurs On	blenderaugerlong-FreeParts Split Line1		
Information			
Time	1. s		
Load Step	1		
Substep	1		
Iteration Number	1		
Integration Point Results			
Display Option		Averaged	
Average Across Bodies		No	

FIGURE 9
Model (A4) > Static Structural (A5) > Solution (A6) > Total Deformation

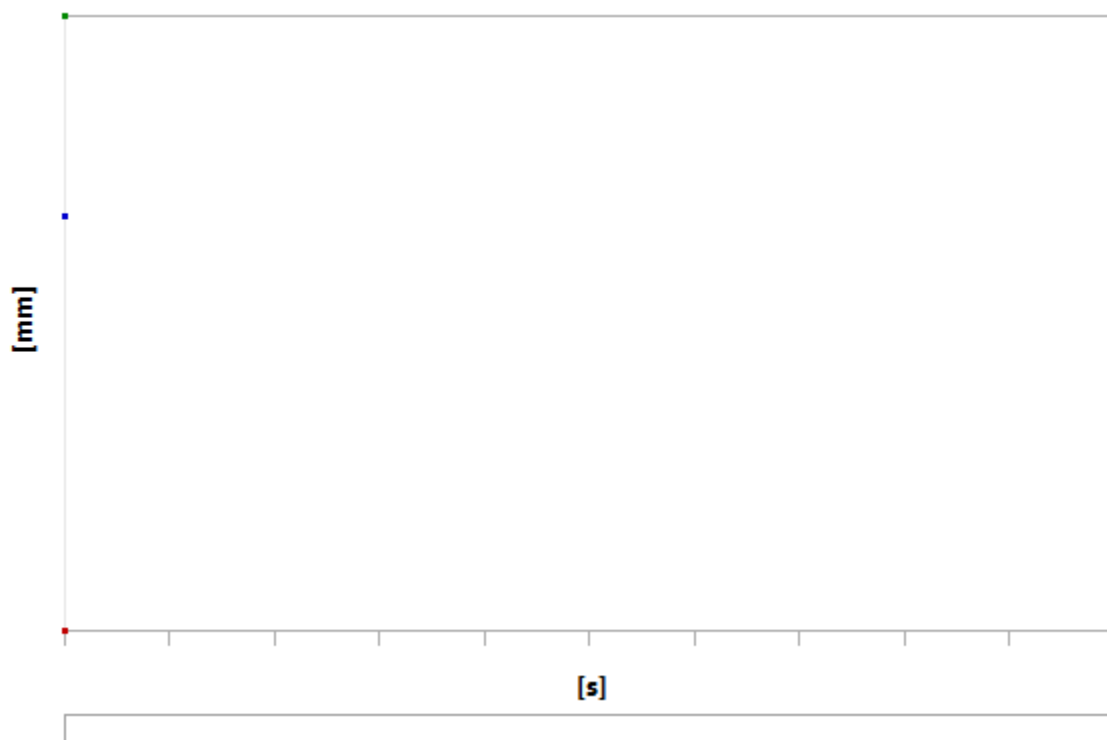


TABLE 16
Model (A4) > Static Structural (A5) > Solution (A6) > Total Deformation

Time [s]	Minimum [mm]	Maximum [mm]	Average [mm]
1.	0.	0.23247	0.15679

FIGURE 10

Model (A4) > Static Structural (A5) > Solution (A6) > Total Deformation > Figure

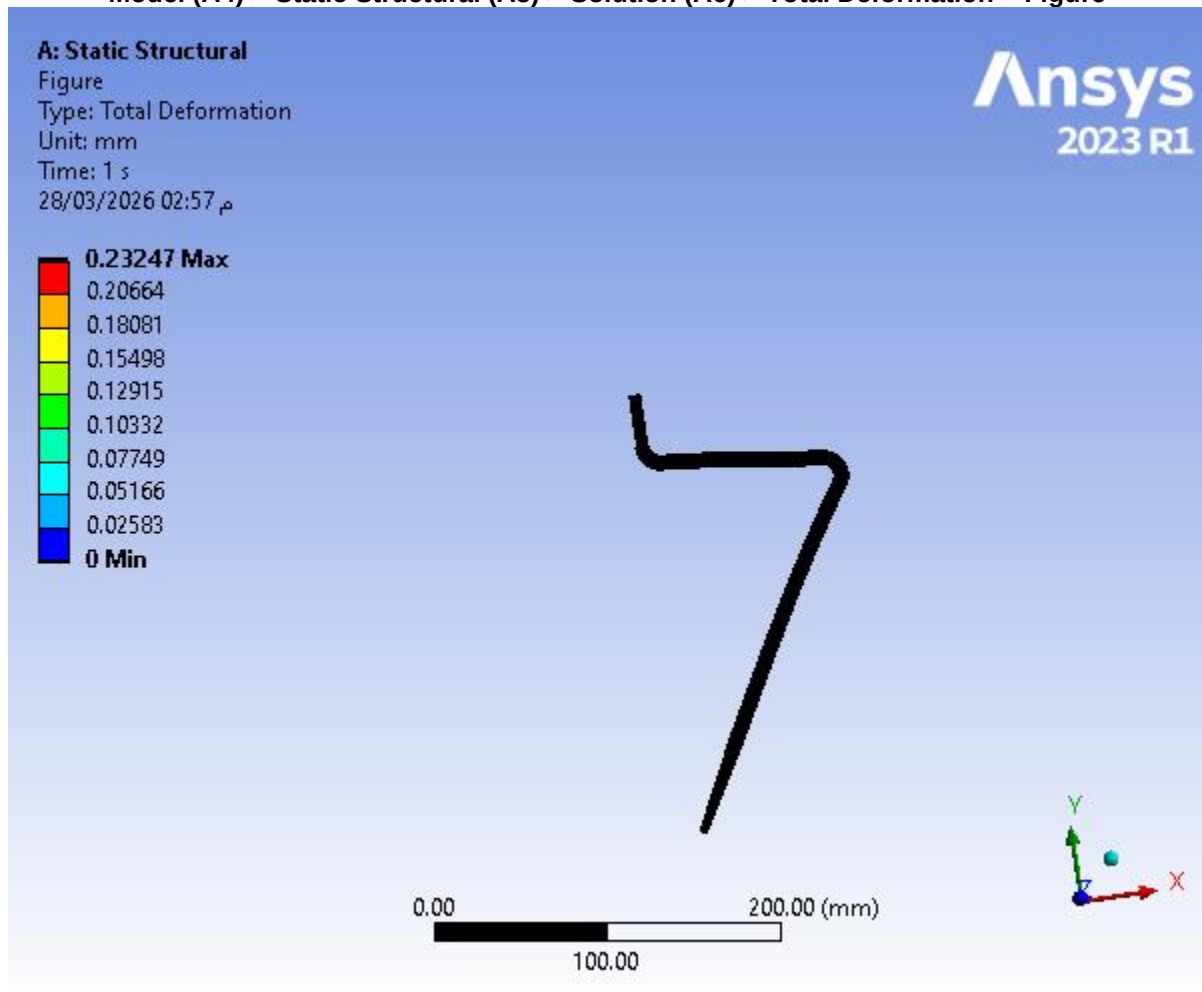


FIGURE 11

Model (A4) > Static Structural (A5) > Solution (A6) > Total Deformation > Image

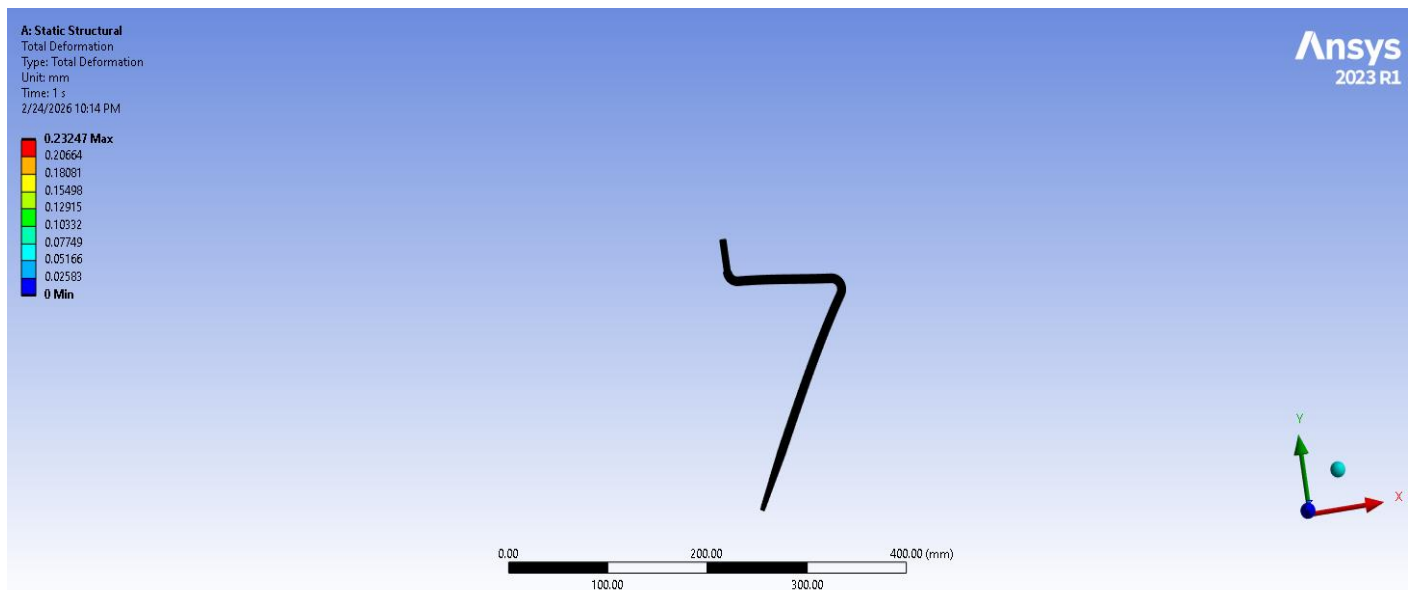


FIGURE 12
 Model (A4) > Static Structural (A5) > Solution (A6) > Maximum Shear Stress

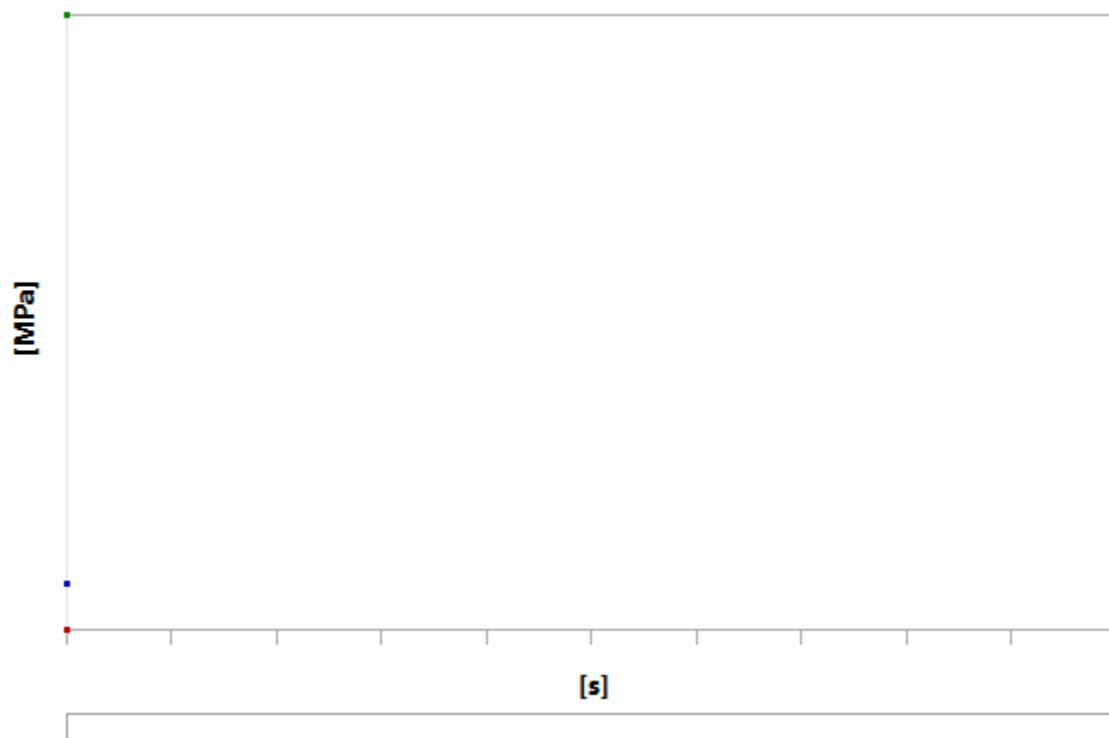


TABLE 17
 Model (A4) > Static Structural (A5) > Solution (A6) > Maximum Shear Stress

Time [s]	Minimum [MPa]	Maximum [MPa]	Average [MPa]
1.	3.992e-003	42.824	3.1889

FIGURE 13
Model (A4) > Static Structural (A5) > Solution (A6) > Maximum Shear Stress > Figure

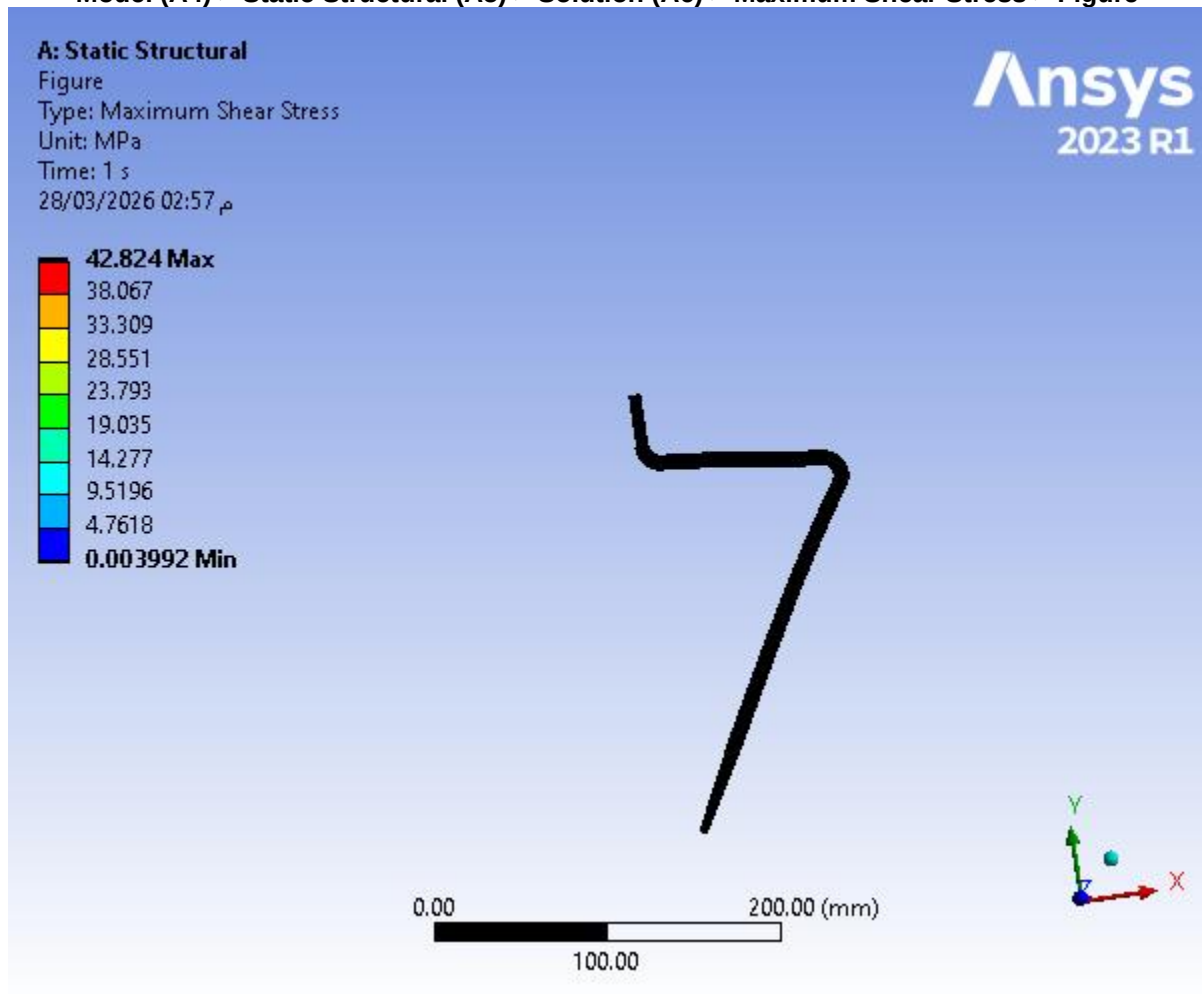


FIGURE 14
Model (A4) > Static Structural (A5) > Solution (A6) > Maximum Shear Stress > Image

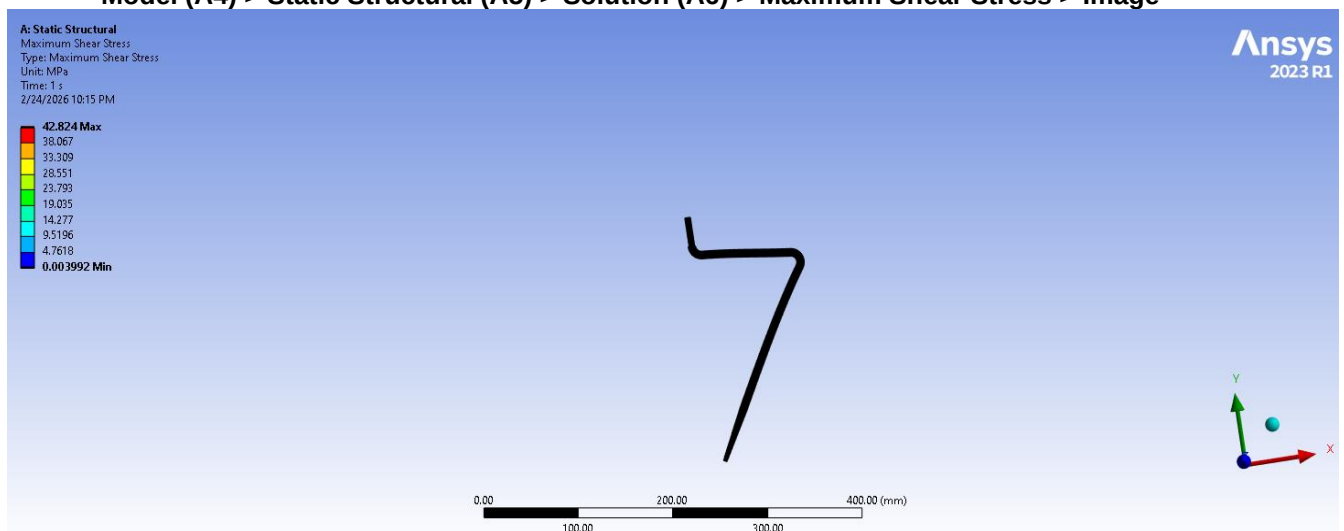


FIGURE 15
Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Stress

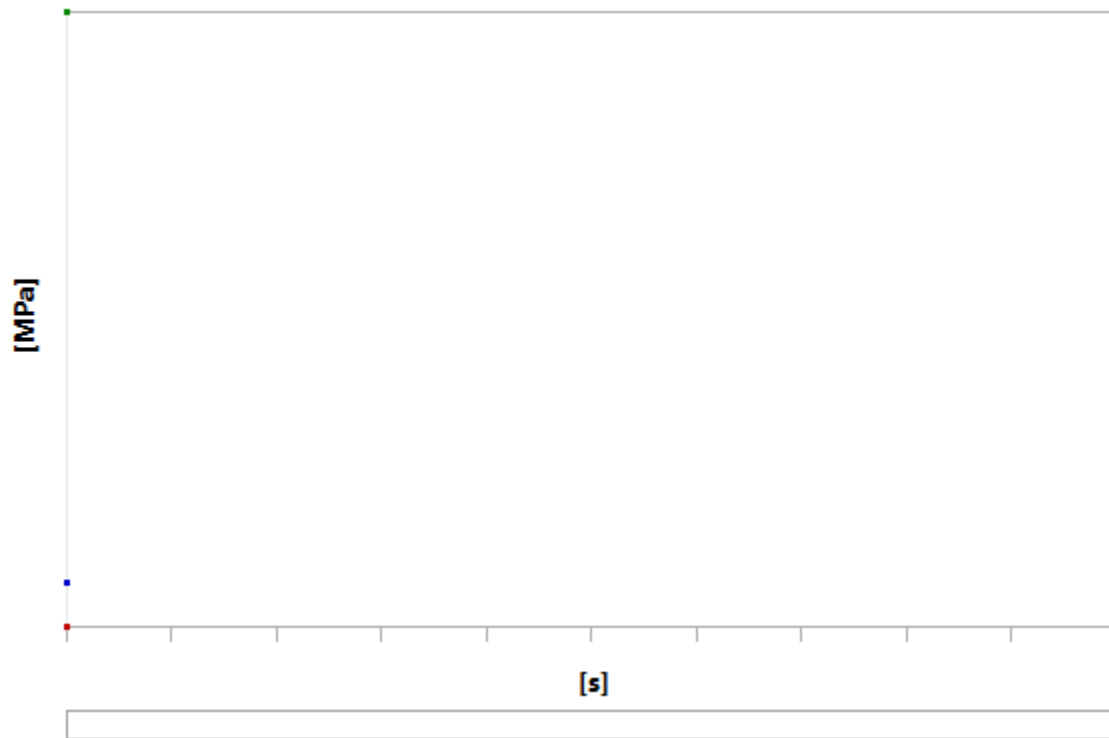


TABLE 18
Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Stress

Time [s]	Minimum [MPa]	Maximum [MPa]	Average [MPa]
1.	7.3206e-003	85.366	6.1837

FIGURE 16
Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Stress > Figure

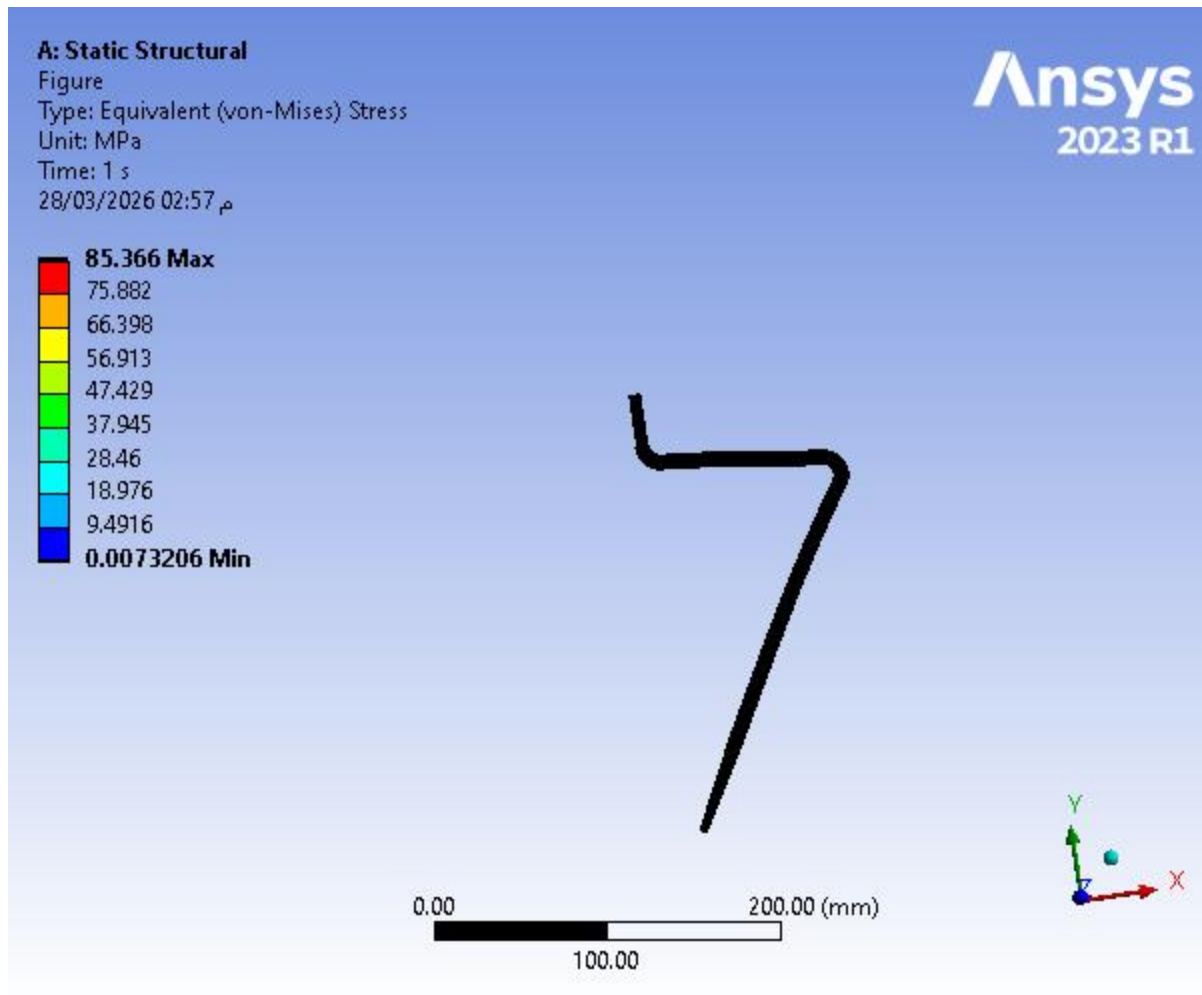


FIGURE 17
Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Stress > Image

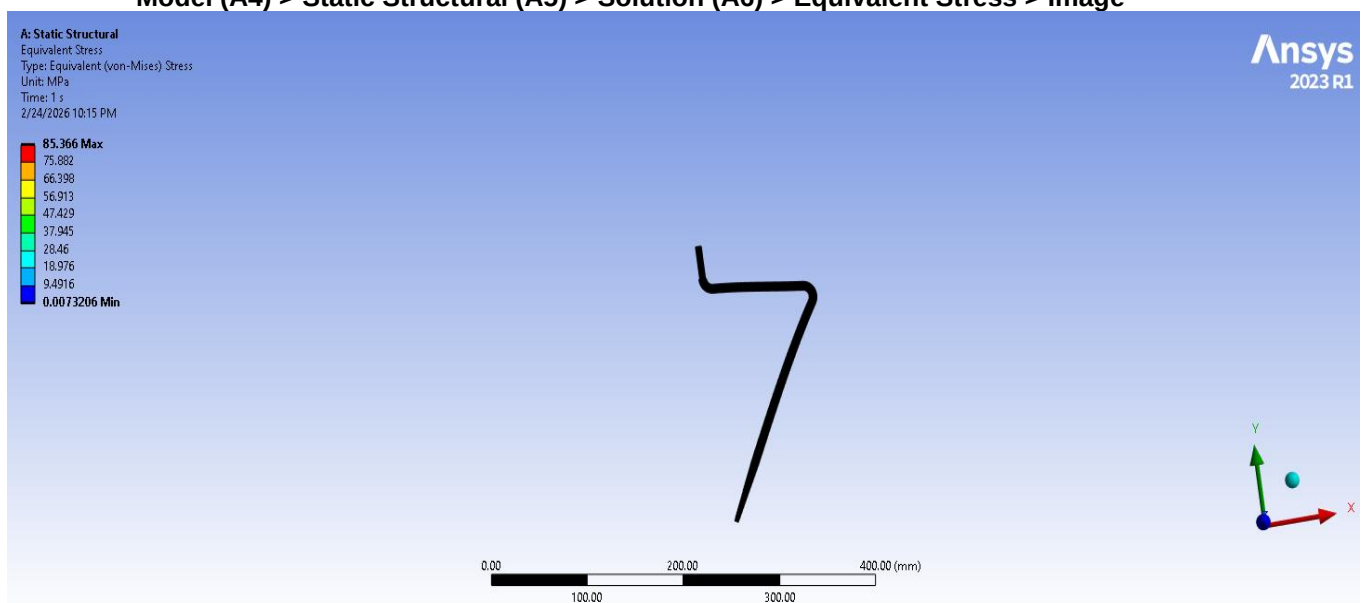


TABLE 19
Model (A4) > Static Structural (A5) > Solution (A6) > Stress Safety Tools

Object Name	<i>Stress Tool</i>
State	Solved
Definition	
Theory	Max Shear Stress
Factor	0.5
Stress Limit Type	Tensile Yield Per Material

TABLE 20
Model (A4) > Static Structural (A5) > Solution (A6) > Stress Tool > Results

Object Name	<i>Safety Factor</i>
State	Solved
Scope	
Scoping Method	Geometry Selection
Geometry	All Bodies
Definition	
Type	Safety Factor
By	Time
Display Time	Last
Separate Data by Entity	No
Calculate Time History	Yes
Identifier	
Suppressed	No
Integration Point Results	
Display Option	Averaged
Average Across Bodies	No
Results	
Minimum	2.9189
Minimum Occurs On	blenderaugerlong-FreeParts Split Line1
Information	
Time	1. s
Load Step	1
Substep	1
Iteration Number	1

FIGURE 18
Model (A4) > Static Structural (A5) > Solution (A6) > Stress Tool > Safety Factor

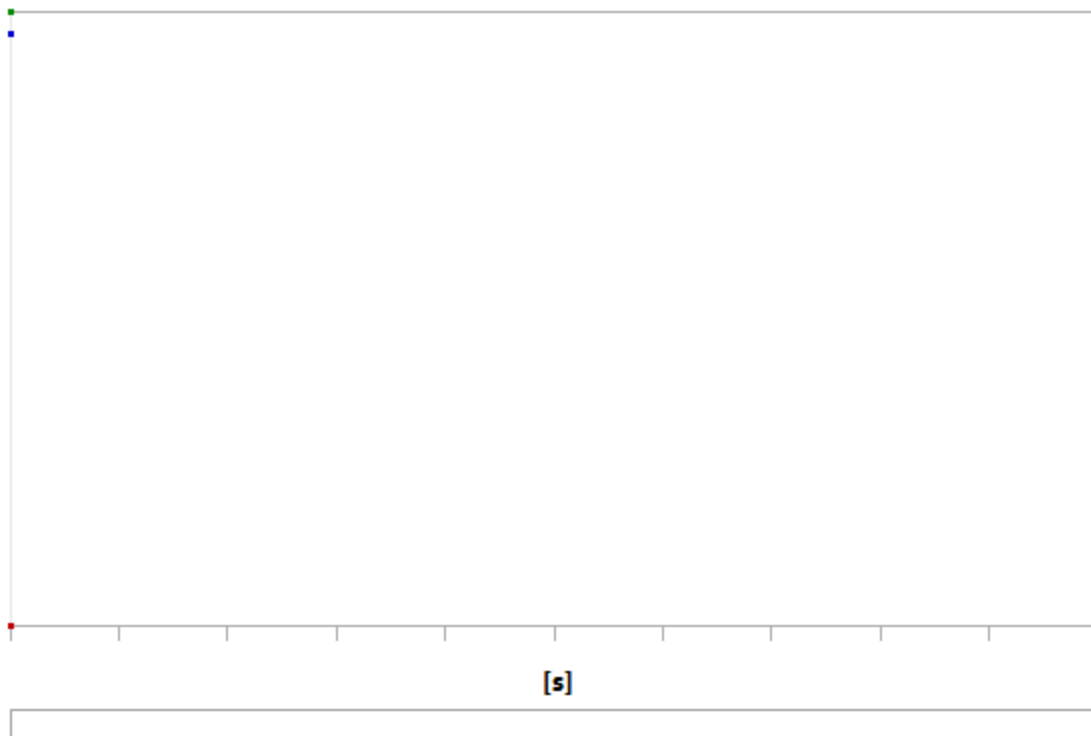


TABLE 21
Model (A4) > Static Structural (A5) > Solution (A6) > Stress Tool > Safety Factor

Time [s]	Minimum	Maximum	Average
1.	2.9189	15.	14.572

FIGURE 19
Model (A4) > Static Structural (A5) > Solution (A6) > Stress Tool > Safety Factor > Figure

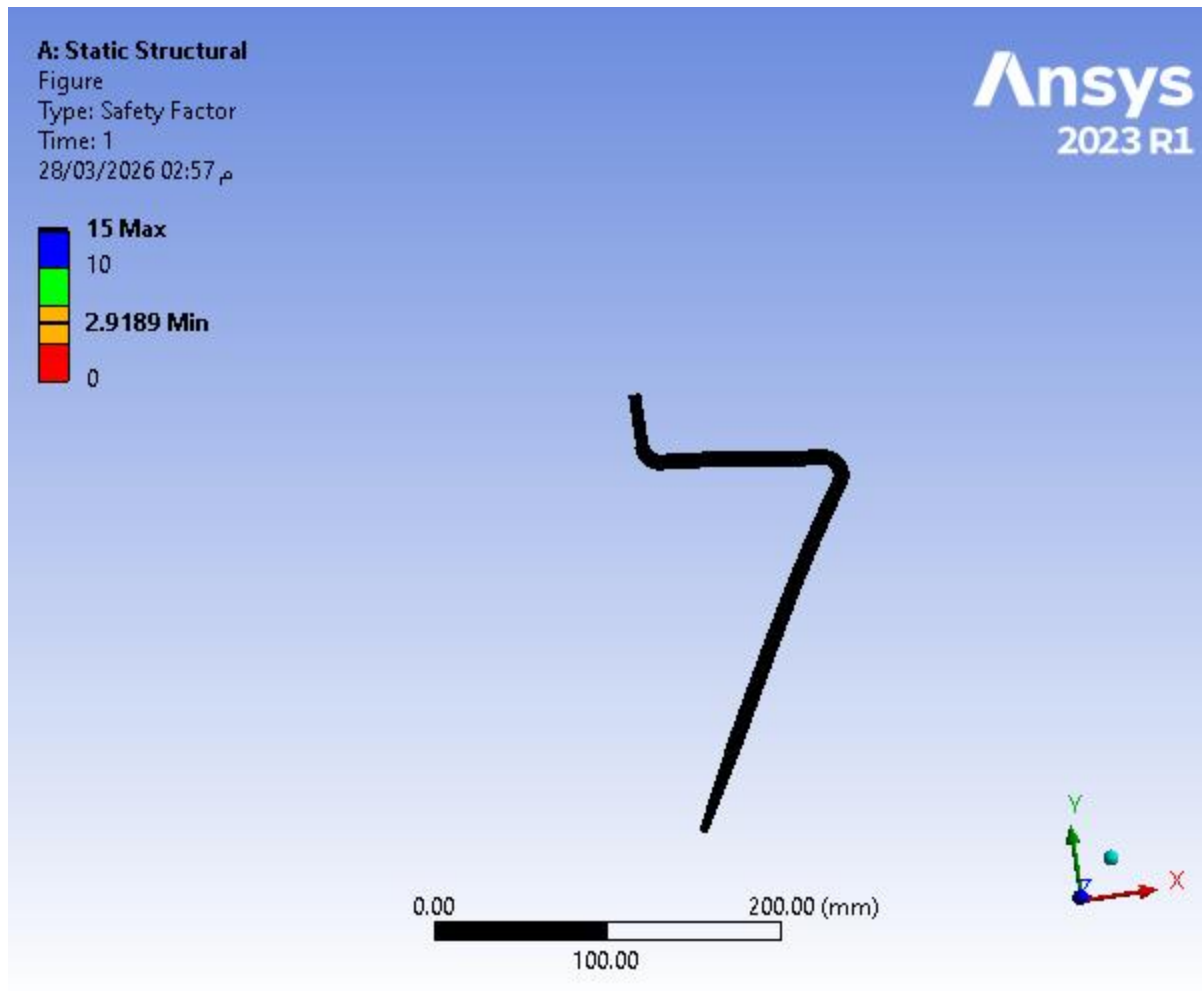


FIGURE 20
Model (A4) > Static Structural (A5) > Solution (A6) > Stress Tool > Safety Factor > Image

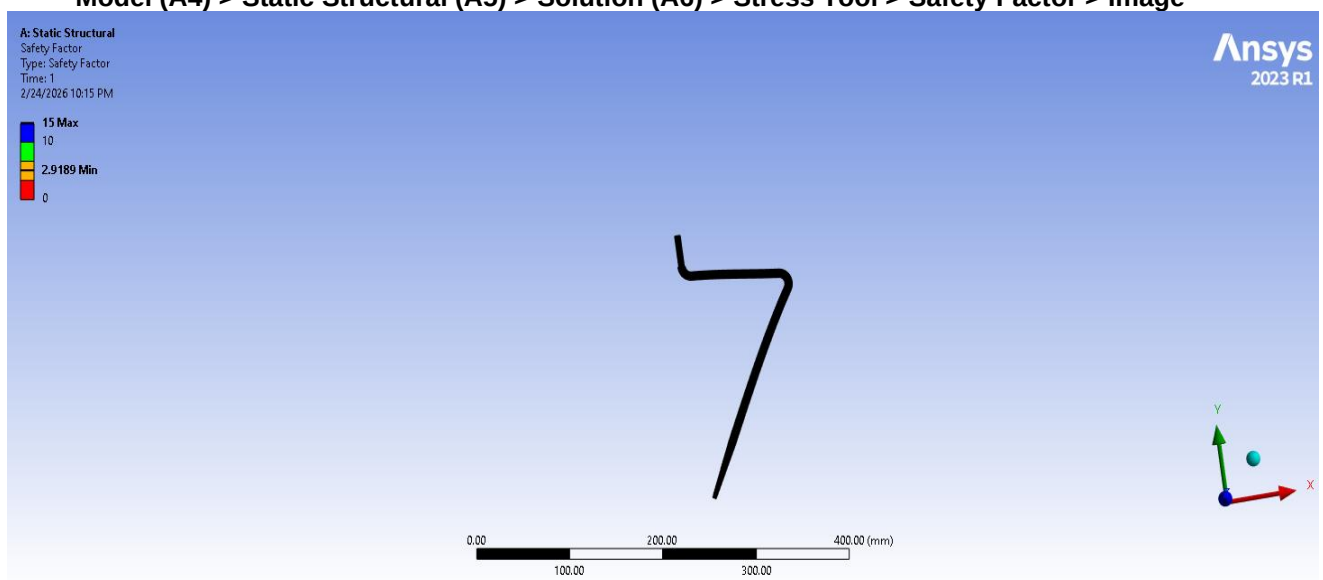


TABLE 22
Model (A4) > Static Structural (A5) > Solution (A6) > Stress Safety Tools

Object Name	<i>Stress Tool 2</i>
State	Solved
Definition	
Theory	Max Equivalent Stress
Stress Limit Type	Tensile Yield Per Material

TABLE 23
Model (A4) > Static Structural (A5) > Solution (A6) > Stress Tool 2 > Results

Object Name	<i>Safety Factor</i>
State	Solved
Scope	
Scoping Method	Geometry Selection
Geometry	All Bodies
Definition	
Type	Safety Factor
By	Time
Display Time	Last
Separate Data by Entity	No
Calculate Time History	Yes
Identifier	
Suppressed	No
Integration Point Results	
Display Option	Averaged
Average Across Bodies	No
Results	
Minimum	2.9286
Minimum Occurs On	blenderaugerlong-FreeParts Split Line1
Information	
Time	1. s
Load Step	1
Substep	1
Iteration Number	1

FIGURE 21
Model (A4) > Static Structural (A5) > Solution (A6) > Stress Tool 2 > Safety Factor

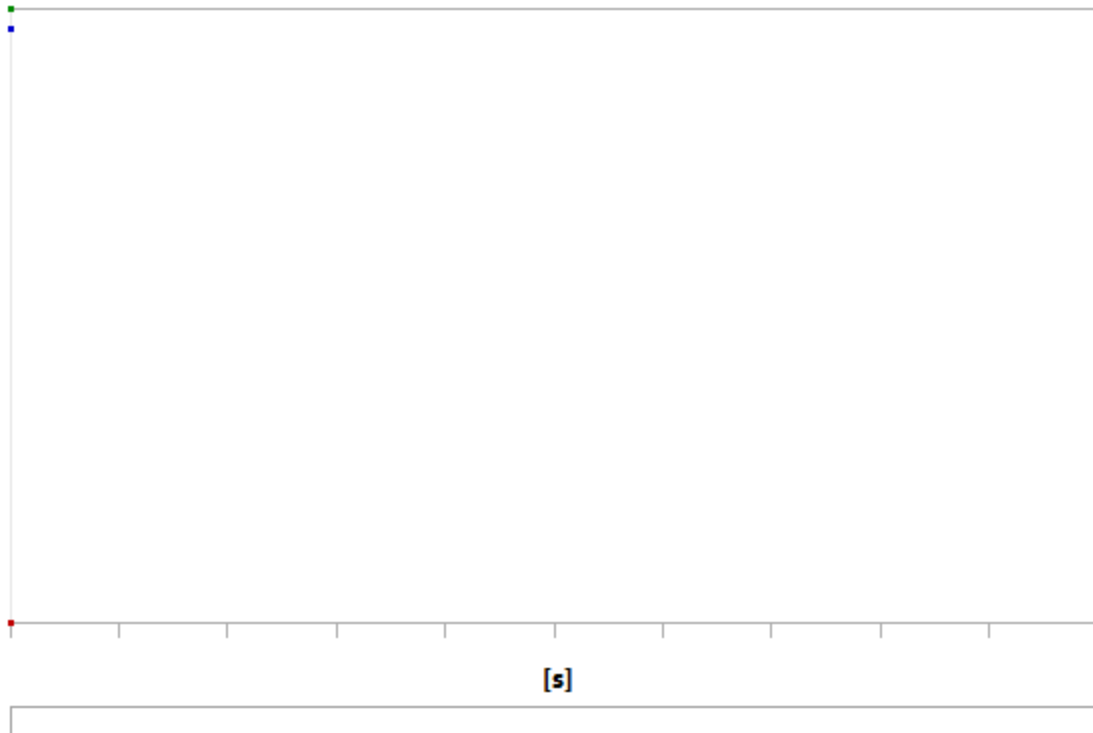


TABLE 24

Model (A4) > Static Structural (A5) > Solution (A6) > Stress Tool 2 > Safety Factor

Time [s]	Minimum	Maximum	Average
1.	2.9286	15.	14.608

FIGURE 22

Model (A4) > Static Structural (A5) > Solution (A6) > Stress Tool 2 > Safety Factor > Figure

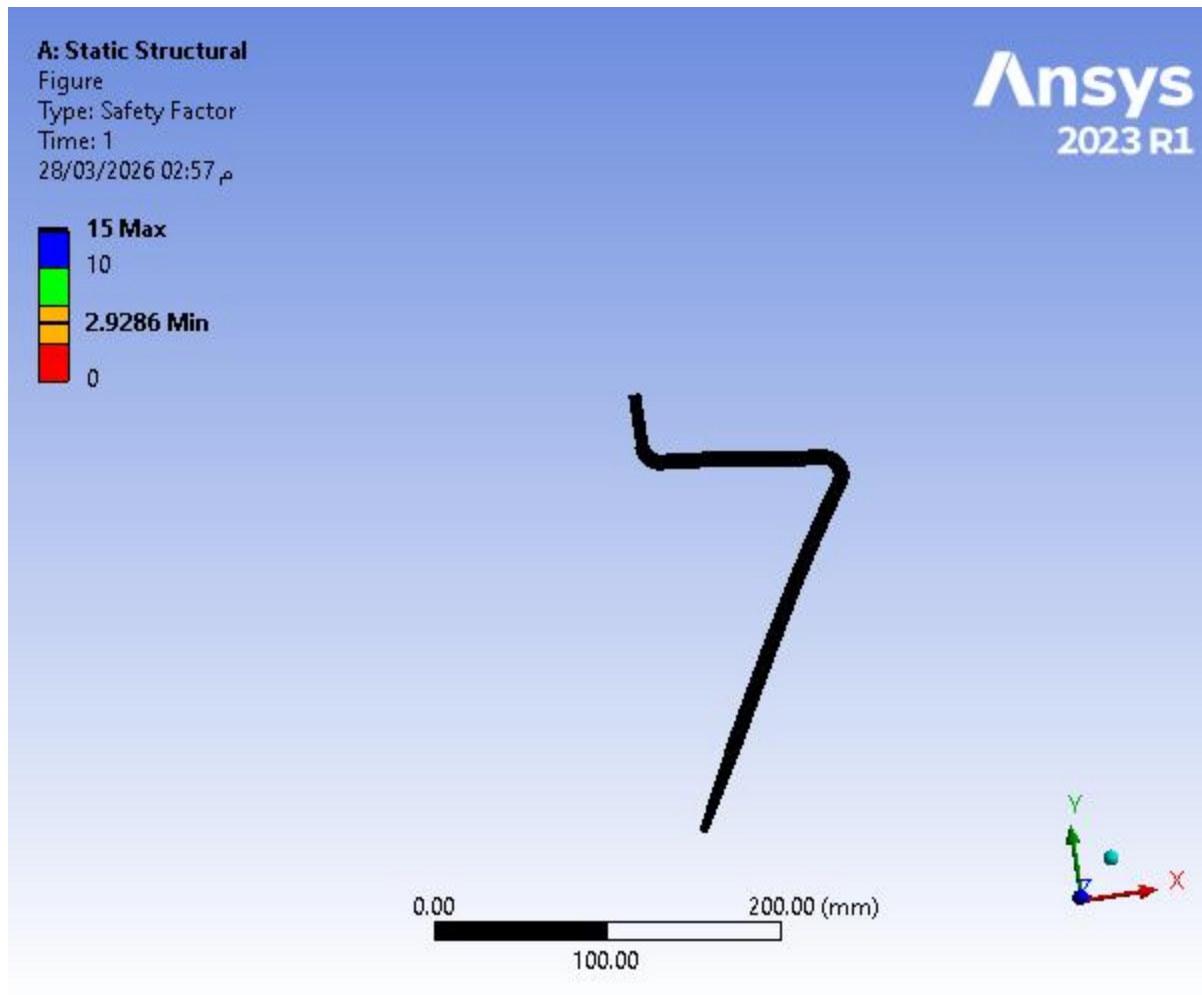


FIGURE 23

Model (A4) > Static Structural (A5) > Solution (A6) > Stress Tool 2 > Safety Factor > Image

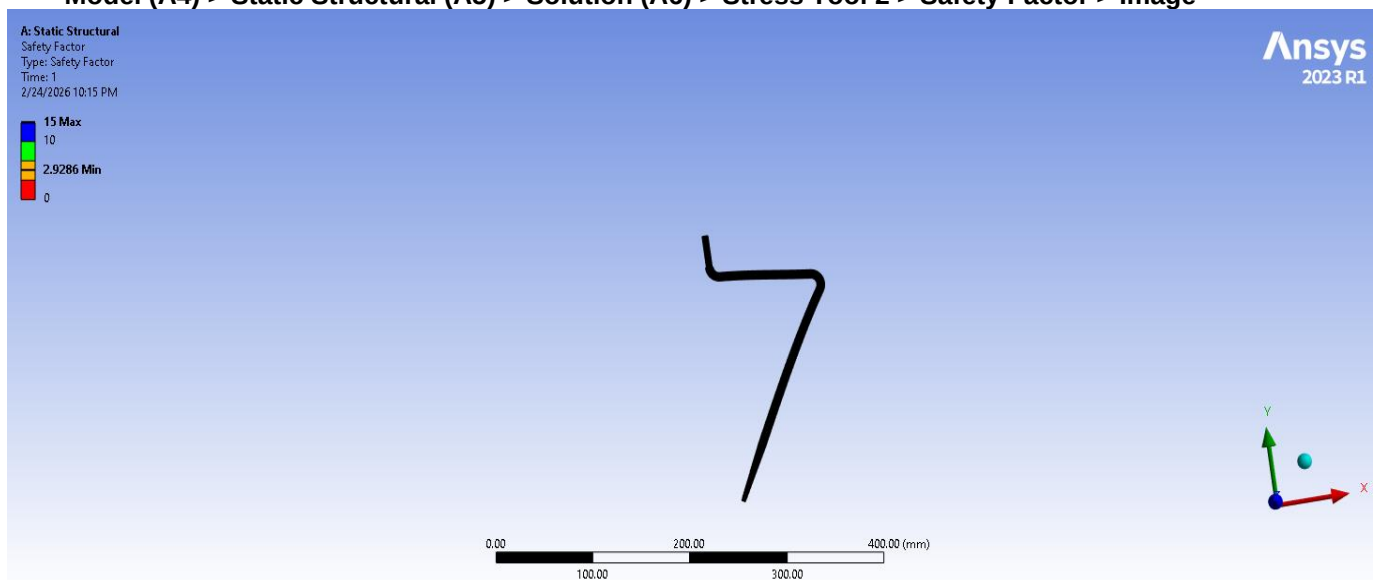


TABLE 25
Model (A4) > Static Structural (A5) > Solution (A6) > Fatigue Tools

Object Name	<i>Fatigue Tool</i>
State	Solved
Domain	
Domain Type	Time
Materials	
Fatigue Strength Factor (Kf)	1.
Loading	
Type	Fully Reversed
Scale Factor	1.
Definition	
Display Time	End Time
Options	
Analysis Type	Stress Life
Mean Stress Theory	Soderberg
Stress Component	Equivalent (von-Mises)
Life Units	
Units Name	cycles
1 cycle is equal to	1. cycles

FIGURE 24
Model (A4) > Static Structural (A5) > Solution (A6) > Fatigue Tool

Constant Amplitude Load
Fully Reversed

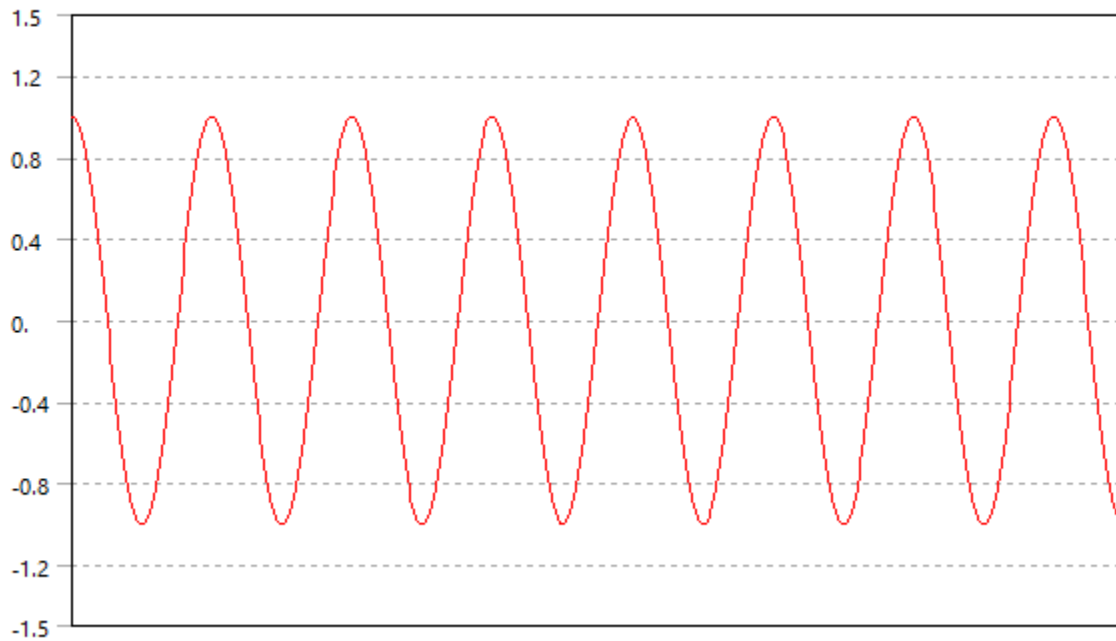


FIGURE 25
Model (A4) > Static Structural (A5) > Solution (A6) > Fatigue Tool

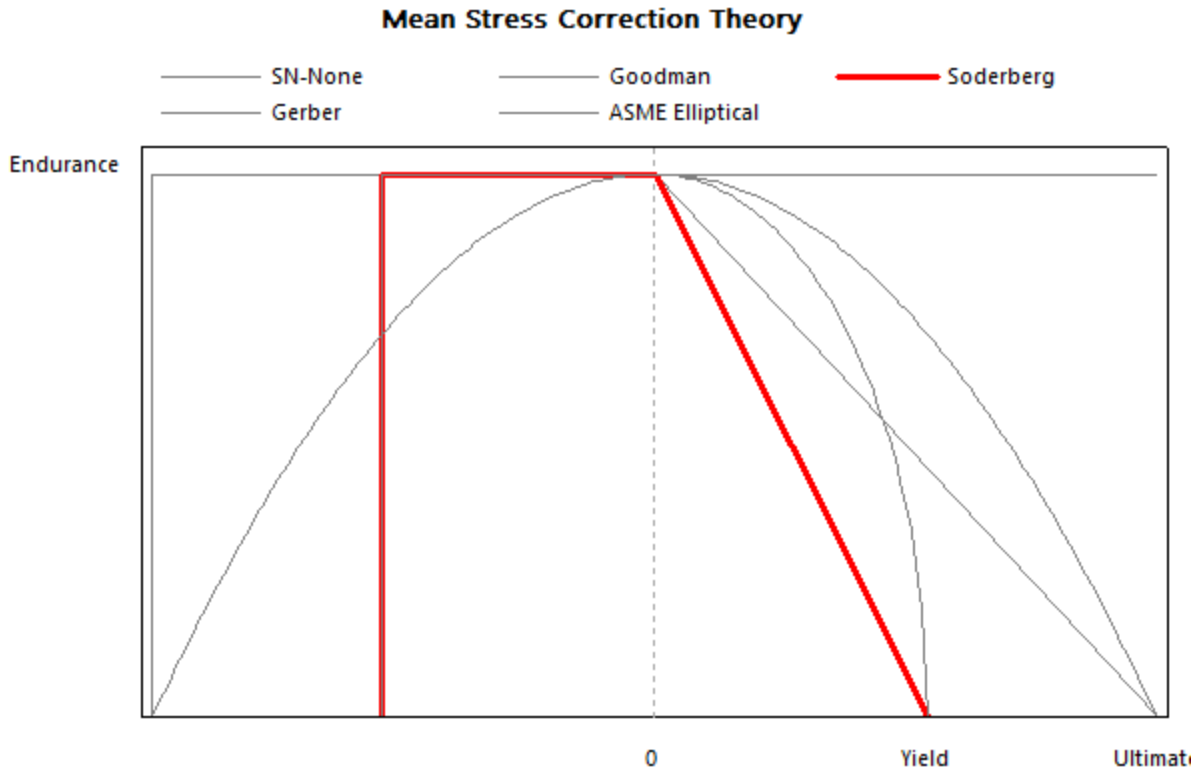


TABLE 26
Model (A4) > Static Structural (A5) > Solution (A6) > Fatigue Tool > Results

Object Name	Life	Safety Factor	Damage
State	Solved		
Scope			
Scoping Method	Geometry Selection		
Geometry	All Bodies		
Definition			
Type	Life	Safety Factor	Damage
Identifier			
Suppressed	No		
Design Life		1.e+009 cycles	
Results			
Minimum	1.e+006 cycles	1.0098	
Minimum Occurs On	blenderaugerlong-FreeParts Split Line1		
Maximum			1000.
Maximum Occurs On			blenderaugerlong-FreeParts Split Line1

FIGURE 26
Model (A4) > Static Structural (A5) > Solution (A6) > Fatigue Tool > Life

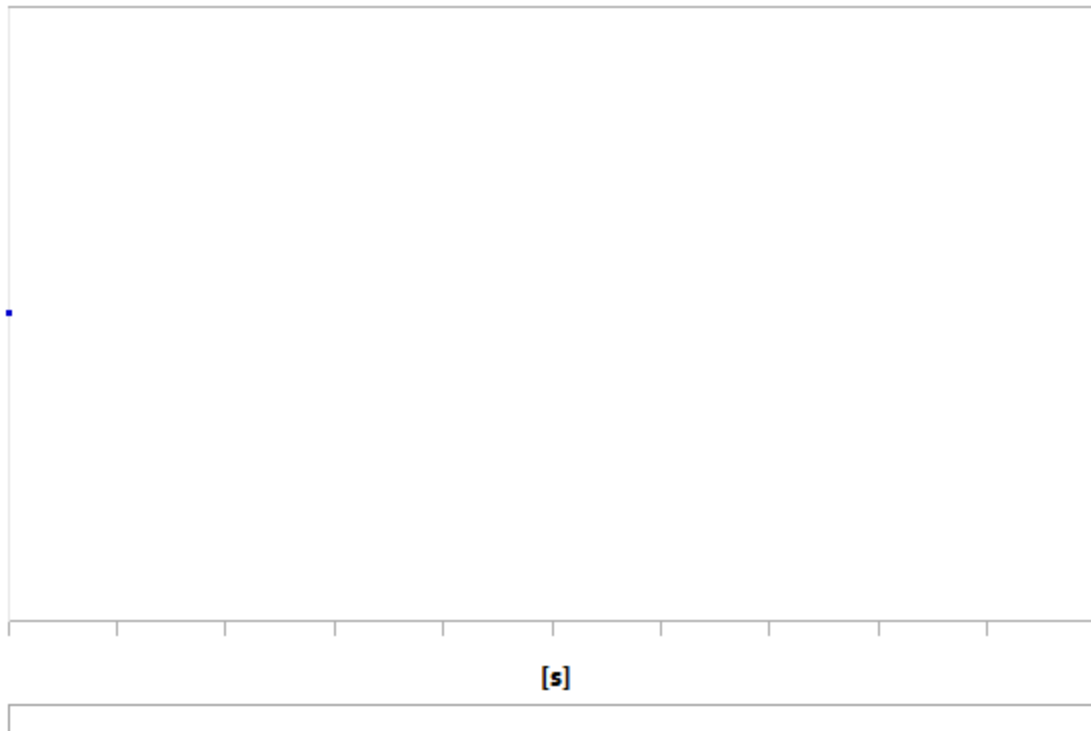


TABLE 27
Model (A4) > Static Structural (A5) > Solution (A6) > Fatigue Tool > Life

Time [s]	Minimum	Maximum	Average
1.	1.e+006	1.e+006	1.e+006

FIGURE 27
Model (A4) > Static Structural (A5) > Solution (A6) > Fatigue Tool > Life > Figure

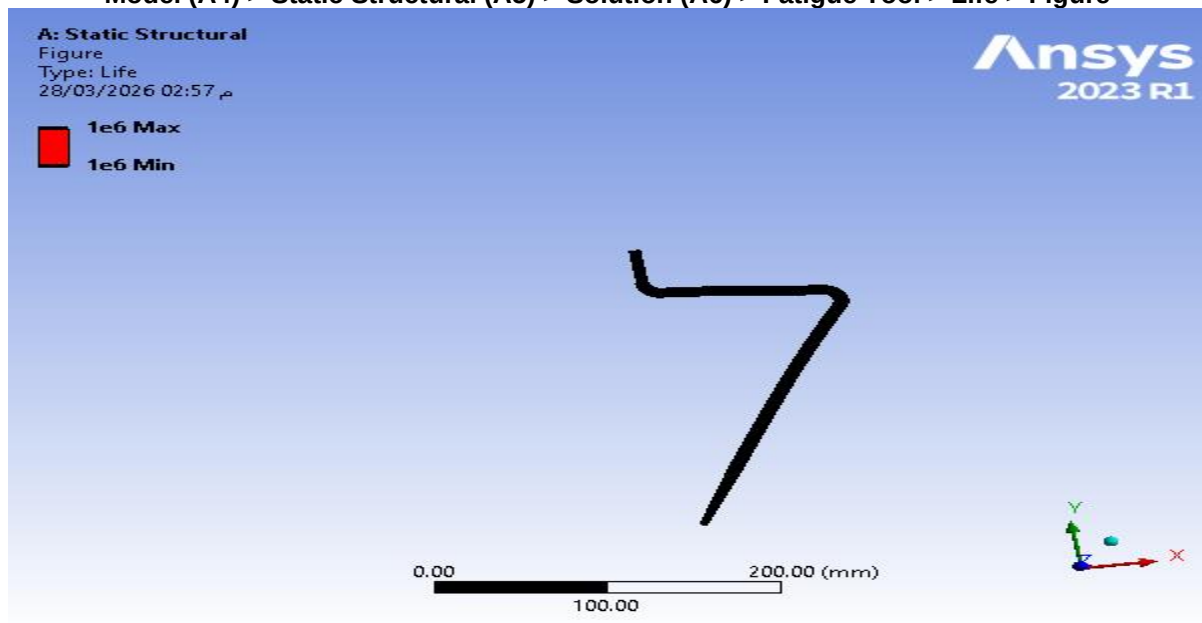


FIGURE 28
Model (A4) > Static Structural (A5) > Solution (A6) > Fatigue Tool > Life > Image

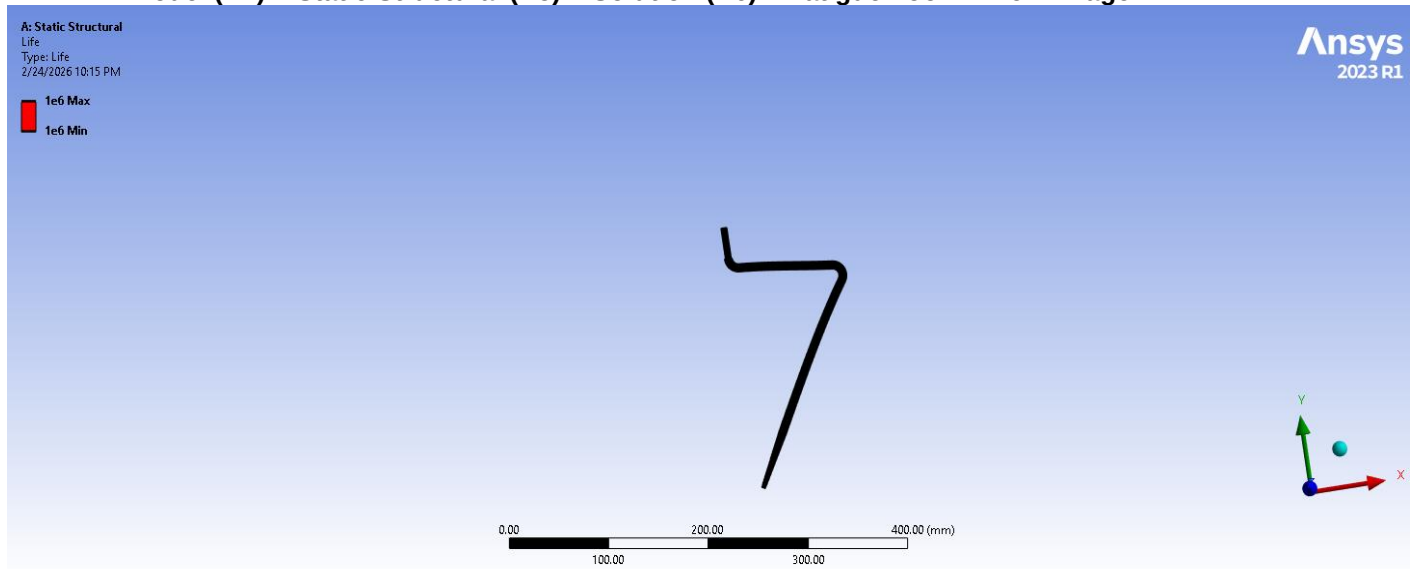


FIGURE 29
Model (A4) > Static Structural (A5) > Solution (A6) > Fatigue Tool > Safety Factor

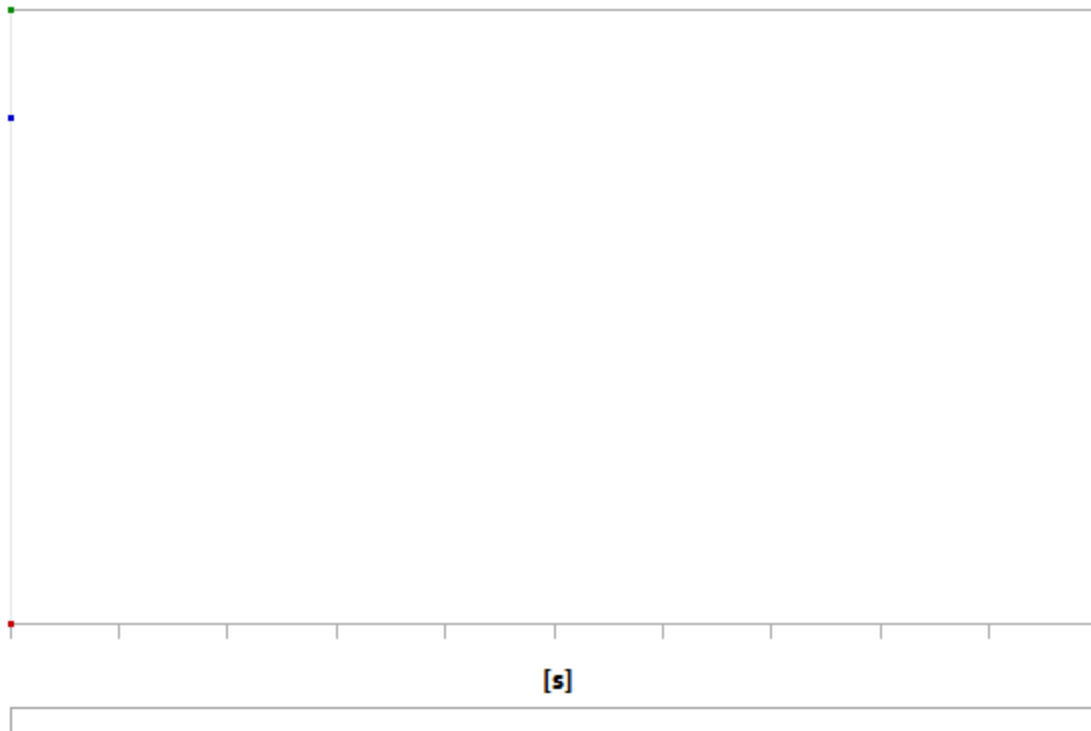


TABLE 28
Model (A4) > Static Structural (A5) > Solution (A6) > Fatigue Tool > Safety Factor

Time [s]	Minimum	Maximum	Average
1.	1.0098	15.	12.535

FIGURE 30

Model (A4) > Static Structural (A5) > Solution (A6) > Fatigue Tool > Safety Factor > Figure

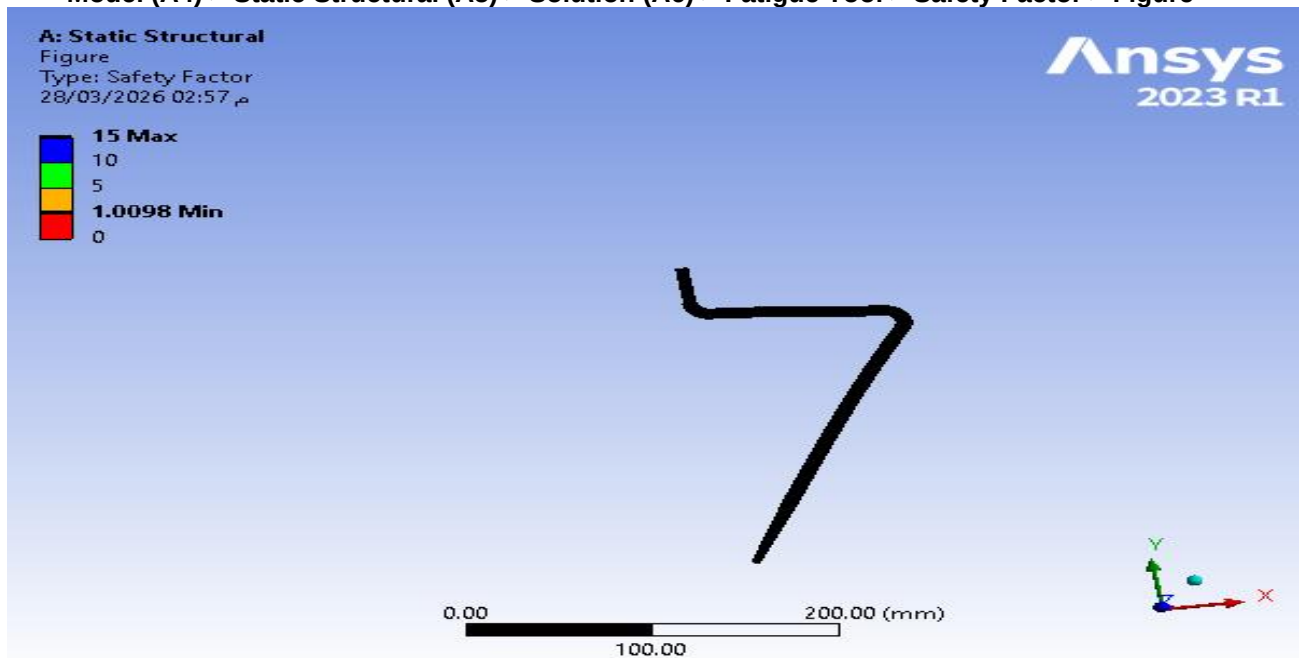


FIGURE 31

Model (A4) > Static Structural (A5) > Solution (A6) > Fatigue Tool > Safety Factor > Image

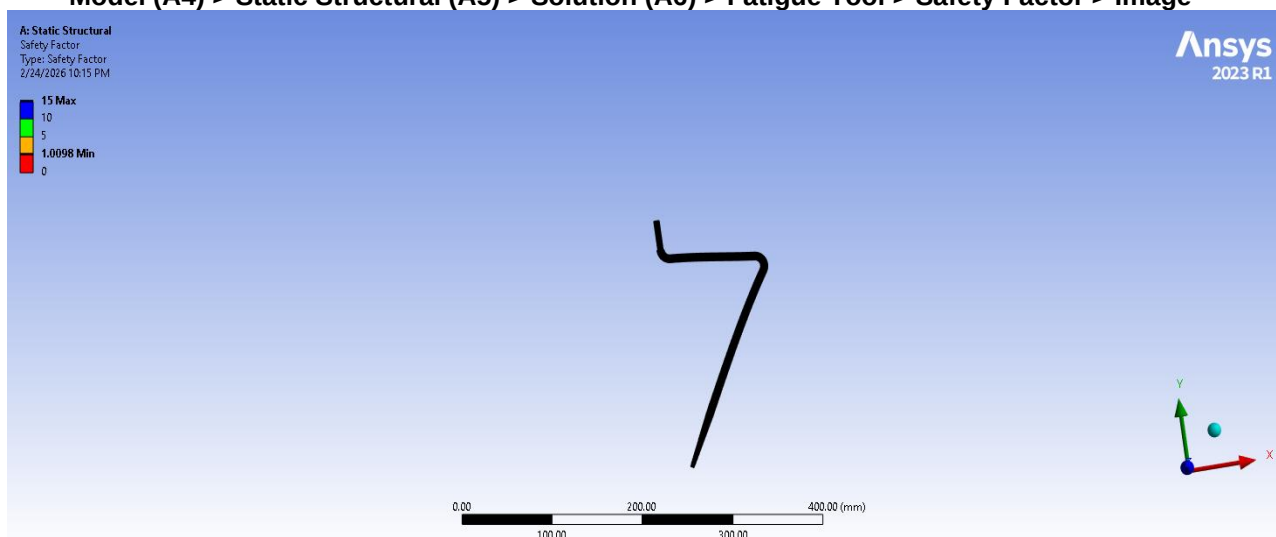


FIGURE 32

Model (A4) > Static Structural (A5) > Solution (A6) > Fatigue Tool > Damage

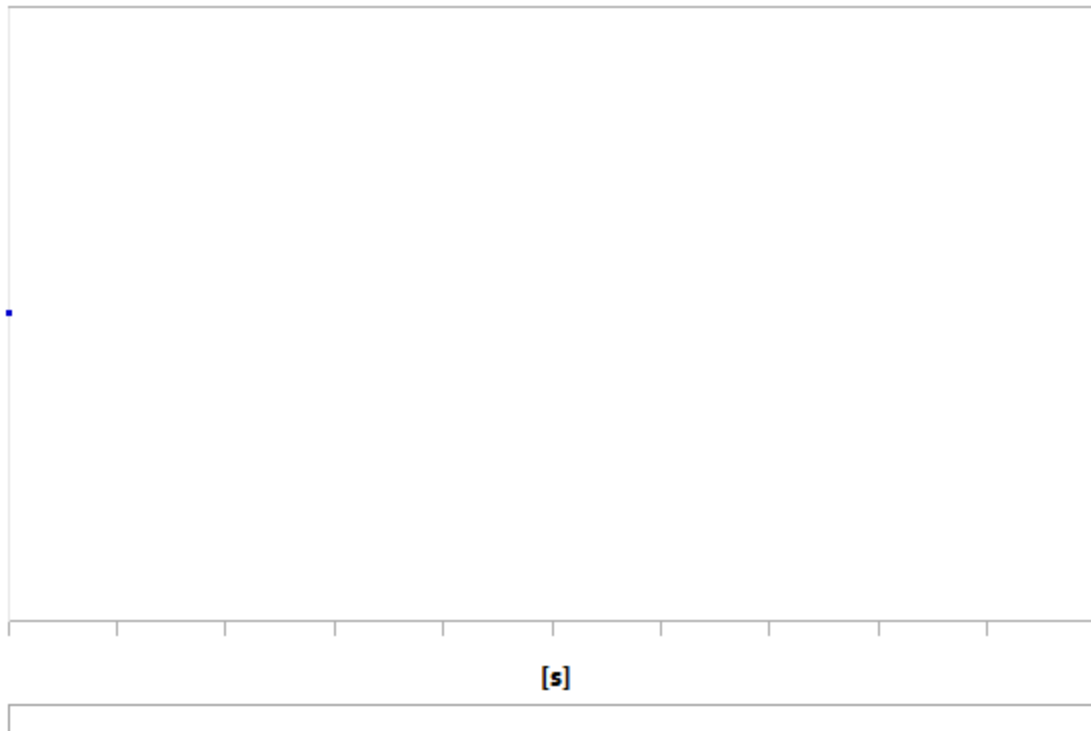


TABLE 29
Model (A4) > Static Structural (A5) > Solution (A6) > Fatigue Tool > Damage

Time [s]	Minimum	Maximum	Average
1.	1000.	1000.	1000.

FIGURE 33
Model (A4) > Static Structural (A5) > Solution (A6) > Fatigue Tool > Damage > Figure

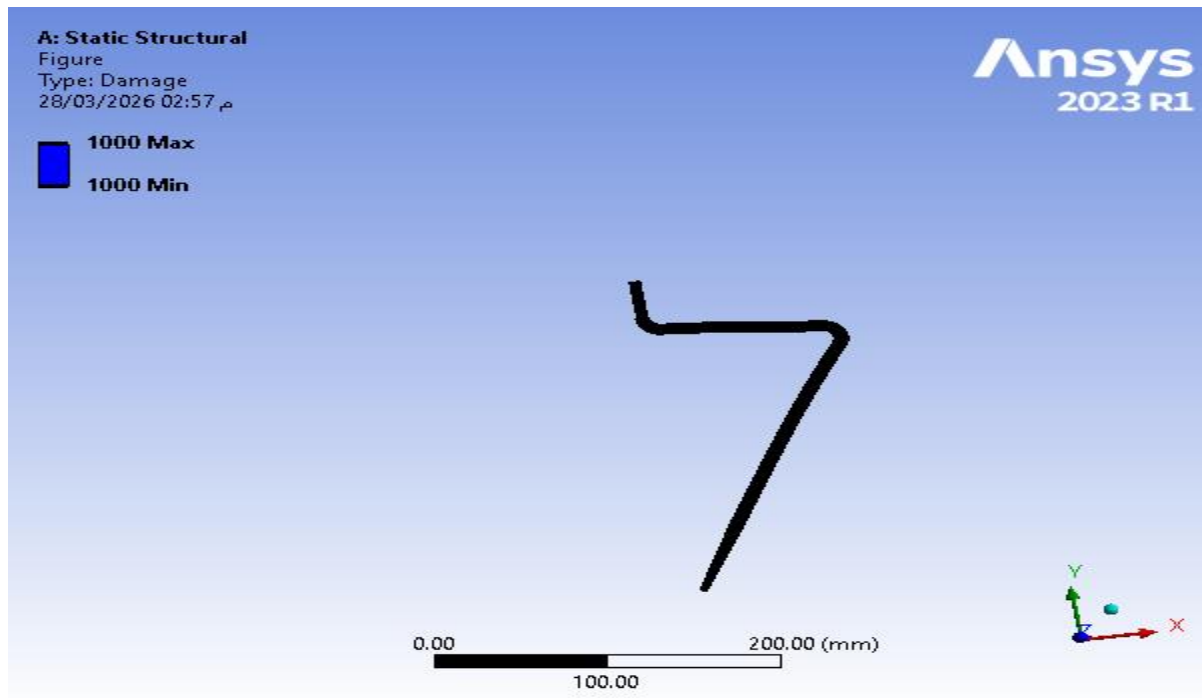
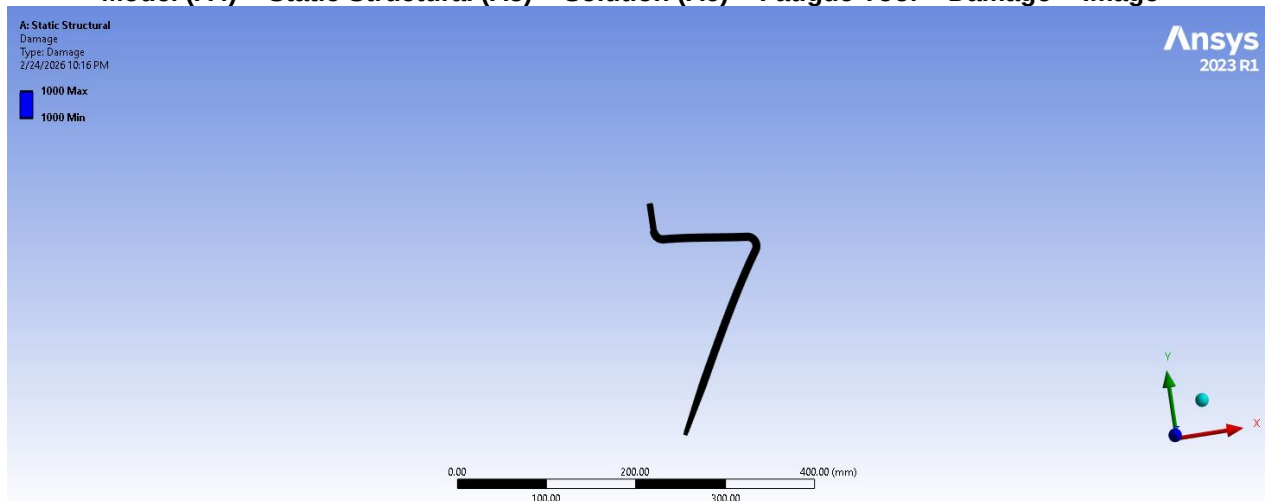


FIGURE 34

Model (A4) > Static Structural (A5) > Solution (A6) > Fatigue Tool > Damage > Image



Material Data

Structural Steel

TABLE 30
Structural Steel > Constants

Density	7.85e-006 kg mm ⁻³
Coefficient of Thermal Expansion	1.2e-005 C ⁻¹
Specific Heat	4.34e+005 mJ kg ⁻¹ C ⁻¹

Thermal Conductivity	6.05e-002 W mm ⁻¹ C ⁻¹
Resistivity	1.7e-004 ohm mm

TABLE 31
Structural Steel > Color

Red	Green	Blue
132	139	179

TABLE 32
Structural Steel > Compressive Ultimate Strength

Compressive Ultimate Strength MPa
0

TABLE 33
Structural Steel > Compressive Yield Strength

Compressive Yield Strength MPa
250

TABLE 34
Structural Steel > Tensile Yield Strength

Tensile Yield Strength MPa
250

TABLE 35
Structural Steel > Tensile Ultimate Strength

Tensile Ultimate Strength MPa
460

TABLE 36
Structural Steel > Isotropic Secant Coefficient of Thermal Expansion

Zero-Thermal-Strain Reference Temperature C
22

TABLE 37
Structural Steel > S-N Curve

Alternating Stress MPa	Cycles	Mean Stress MPa
3999	10	0
2827	20	0
1896	50	0
1413	100	0
1069	200	0
441	2000	0
262	10000	0
214	20000	0
138	1.e+005	0
114	2.e+005	0
86.2	1.e+006	0

TABLE 38
Structural Steel > Strain-Life Parameters

Strength Coefficient MPa	Strength Exponent	Ductility Coefficient	Ductility Exponent	Cyclic Strength Coefficient MPa	Cyclic Strain Hardening Exponent
920	-0.106	0.213	-0.47	1000	0.2

TABLE 39
Structural Steel > Isotropic Elasticity

Young's Modulus MPa	Poisson's Ratio	Bulk Modulus MPa	Shear Modulus MPa	Temperature C
2.e+005	0.3	1.6667e+005	76923	

TABLE 40
Structural Steel > Isotropic Relative Permeability

Relative Permeability
10000